

100 DFA Construction Practice Questions — Full Solutions

Every DFA below was built programmatically (KMP-based substring automata, modular counters, product construction for combined conditions, and Hopcroft-style minimization), then verified by brute-force testing against an independent reference definition of each language on all strings up to length 11 over $\Sigma = \{0, 1\}$. Every diagram shown is the **minimal** DFA.

Legend: circle = state, double circle = accepting state, arrow from nowhere = start state.

Contents

Level 1: Basic DFA

1. All strings ending with 0.
2. All strings ending with 1.
3. All strings starting with 0.
4. All strings starting with 1.
5. Strings containing at least one 0.
6. Strings containing at least one 1.
7. Strings containing exactly one 0.
8. Strings containing exactly one 1.
9. Strings containing no 0.
10. Strings containing no 1.

Level 2: Even/Odd Occurrences

11. Even number of 0s.
12. Odd number of 0s.
13. Even number of 1s.
14. Odd number of 1s.
15. Even number of both 0s and 1s.
16. Odd number of both 0s and 1s.
17. Even number of 0s and odd number of 1s.
18. Odd number of 0s and even number of 1s.

Level 3: Counting

19. Exactly two 0s.
20. Exactly three 1s.
21. At least two 0s.
22. At most two 0s.
23. At least three 1s.
24. At most three 1s.
25. Exactly two 0s and one 1.
26. More than two 1s.
27. Fewer than three 0s.
28. Exactly four 0s.

Level 4: Prefix/Suffix

29. Ending with 01.
30. Ending with 10.
31. Ending with 00.
32. Ending with 11.
33. Ending with 001.
34. Ending with 101.
35. Starting with 01.
36. Starting with 10.
37. Starting with 001.
38. Starting with 111.

Level 5: Substrings

39. Containing substring 01.
40. Containing substring 10.
41. Containing substring 101.

- 42. Containing substring 110.
- 43. Containing substring 001.
- 44. Containing substring 111.
- 45. Containing substring 000.
- 46. Containing substring 0101.

Level 6: Avoid Patterns

- 47. Do not contain 11.
- 48. Do not contain 00.
- 49. Do not contain 101.
- 50. Do not contain 010.
- 51. No consecutive 0s.
- 52. No consecutive 1s.
- 53. Do not contain 111.
- 54. Do not contain 001.

Level 7: Length

- 55. Even length.
- 56. Odd length.
- 57. Length divisible by 3.
- 58. Length divisible by 4.
- 59. Length greater than 3.
- 60. Length ≤ 5 .
- 61. Length exactly 4.
- 62. Length at least 5.

Level 8: Modulo

- 63. Binary numbers divisible by 2.
- 64. Binary numbers divisible by 3.
- 65. Binary numbers divisible by 4.
- 66. Binary numbers divisible by 5.
- 67. Binary numbers divisible by 6.
- 68. Binary numbers divisible by 7.

Level 9: Combination

- 69. Ending with 01 and even number of 0s.
- 70. Starting with 1 and ending with 0.
- 71. Even number of 1s and contains 00.
- 72. Starting with 01 and exactly two 1s.
- 73. Ending with 10 and odd number of 0s.
- 74. Beginning with 1 and length divisible by 3.
- 75. Exactly two 0s and ending with 1.
- 76. Contains 101 and ends with 0.

Level 10: Advanced

- 77. Every 0 is immediately followed by 1.
- 78. Every 1 is immediately preceded by 0.
- 79. No two consecutive 1s.
- 80. No three consecutive 0s.
- 81. Third symbol from end is 1.
- 82. Second symbol from end is 0.
- 83. Fourth symbol from end is 1.
- 84. Number of 0s divisible by 3.
- 85. Number of 1s divisible by 4.

86. Both number of 0s and length even.
87. Starts and ends with same symbol.
88. Starts with 01 and ends with 10.
89. Exactly one occurrence of 101.
90. At least two occurrences of 01.
91. Odd occurrences of 11.
92. Does not contain 000 or 111.
93. Ends with 0011.
94. Begins with 1100.
95. Contains both 101 and 010 (as substrings).
96. Exactly one pair of consecutive 1s.
97. Exactly one pair of consecutive 0s.
98. At least one 0 and one 1.
99. Every 00 is followed later by 11.
100. All strings except epsilon.

Question 1

Language: All strings ending with 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

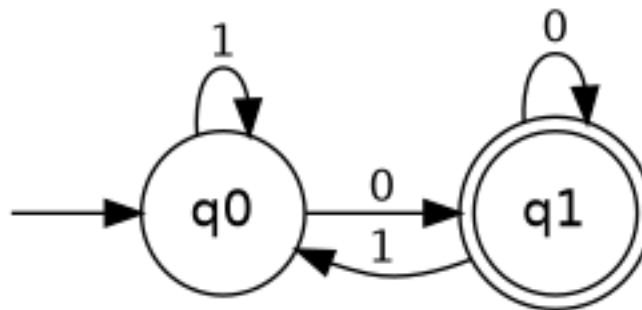
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	
q1	q1	q0		✓

State diagram (minimal DFA)



Question 2

Language: All strings ending with 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

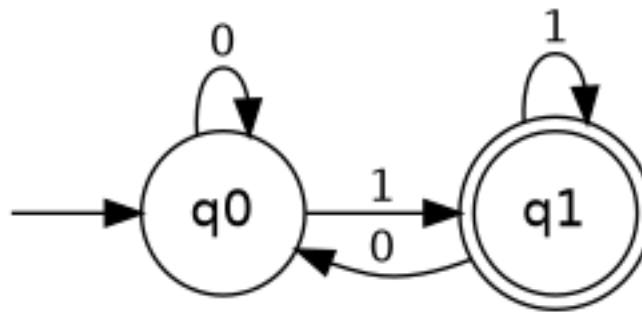
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	
q1	q0	q1		✓

State diagram (minimal DFA)



Question 3

Language: All strings starting with 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

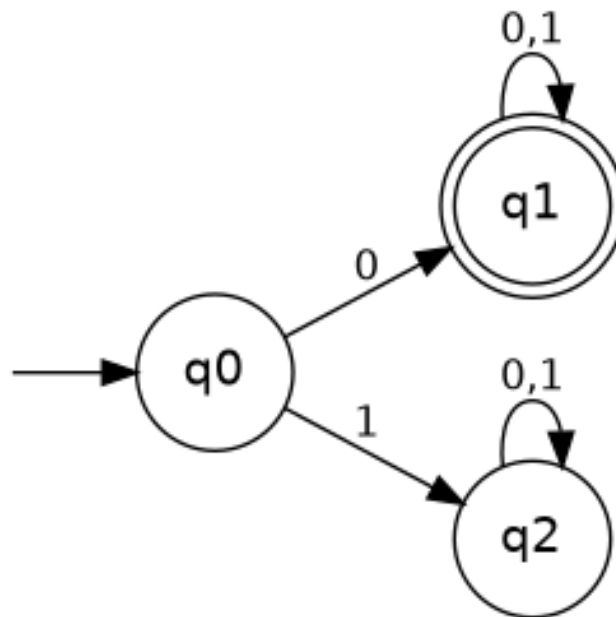
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q2	→	
q1	q1	q1		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 4

Language: All strings starting with 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

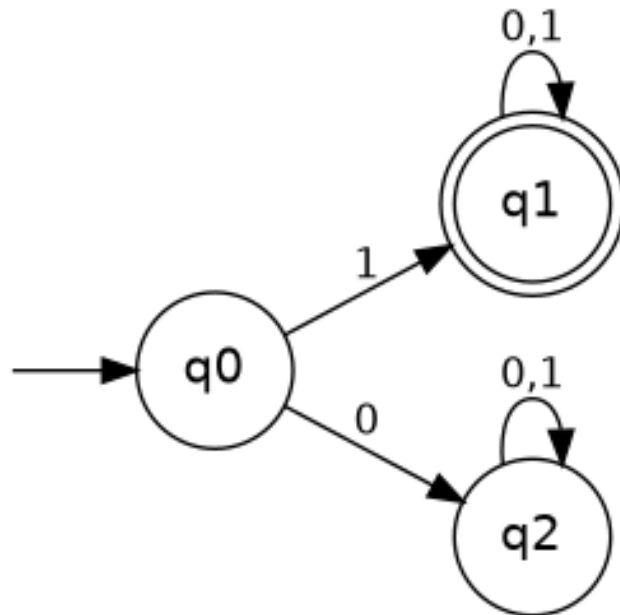
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q1	→	
q1	q1	q1		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 5

Language: Strings containing at least one 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

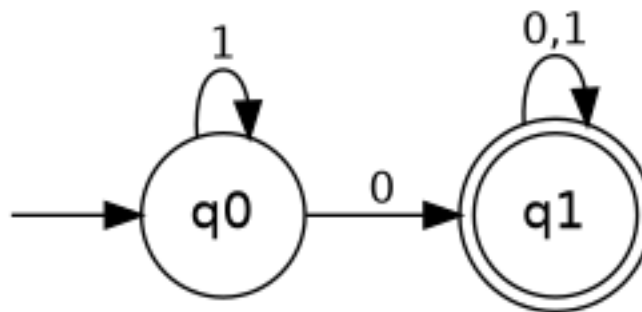
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	
q1	q1	q1		✓

State diagram (minimal DFA)



Question 6

Language: Strings containing at least one 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

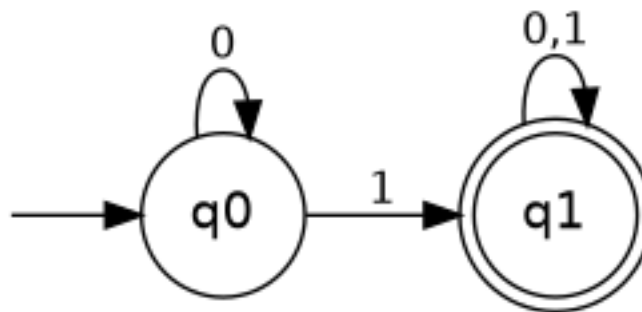
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	
q1	q1	q1		✓

State diagram (minimal DFA)



Question 7

Language: Strings containing exactly one 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

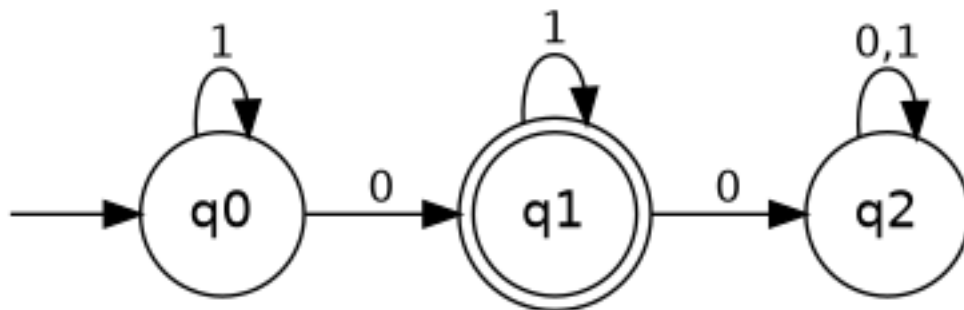
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	
q1	q2	q1		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 8

Language: Strings containing exactly one 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

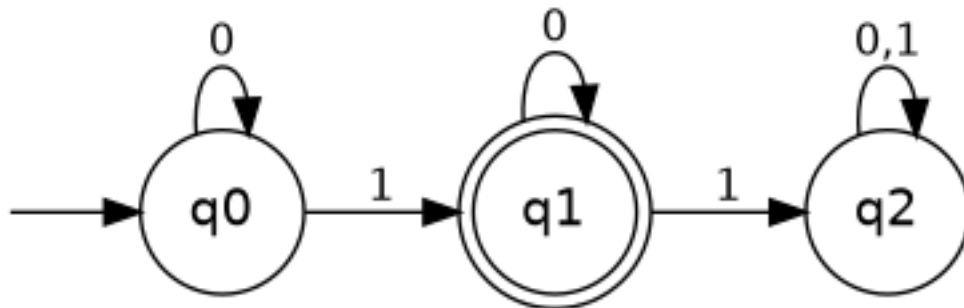
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	
q1	q1	q2		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 9

Language: Strings containing no 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

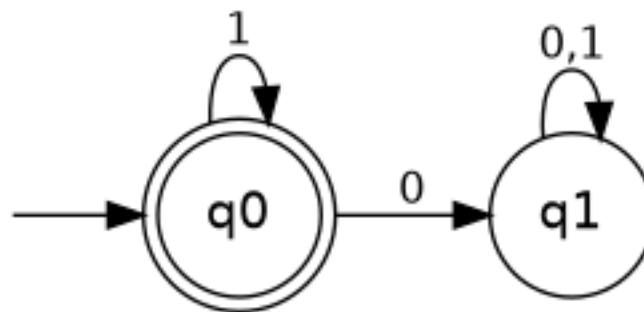
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q1	q1		

State diagram (minimal DFA)



Question 10

Language: Strings containing no 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

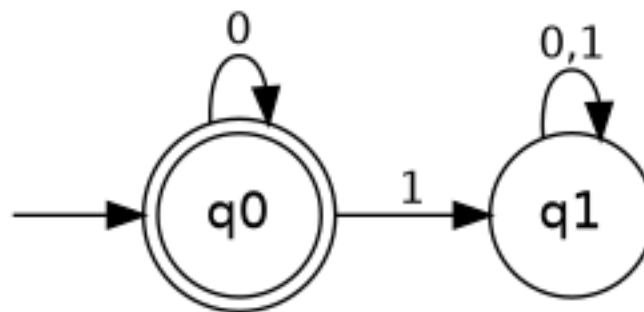
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q1	q1		

State diagram (minimal DFA)



Question 11

Language: Even number of 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

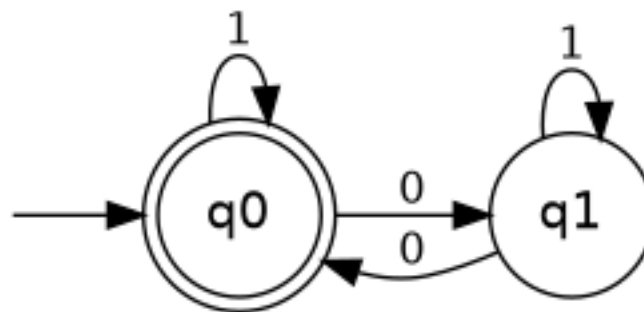
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q0	q1		

State diagram (minimal DFA)



Question 12

Language: Odd number of 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

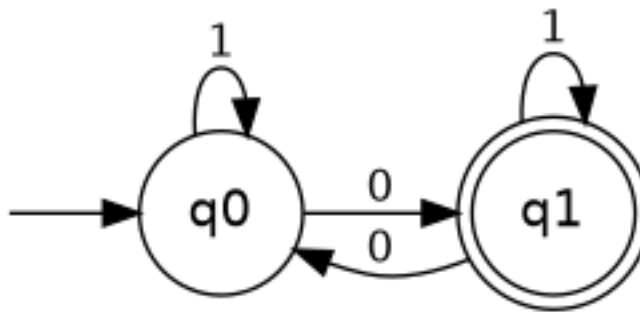
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	
q1	q0	q1		✓

State diagram (minimal DFA)



Question 13

Language: Even number of 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

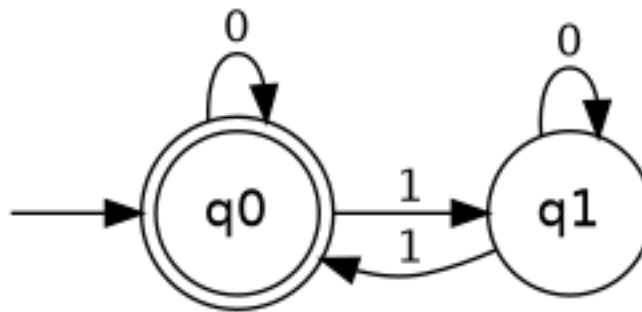
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q1	q0		

State diagram (minimal DFA)



Question 14

Language: Odd number of 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

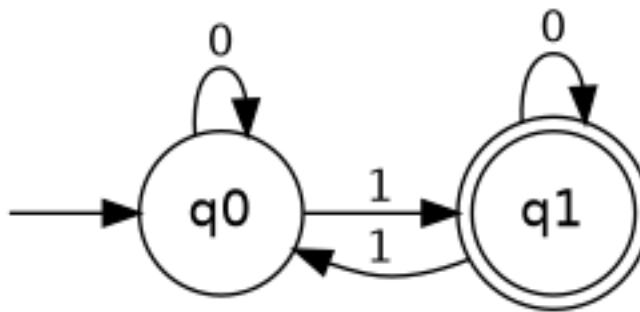
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	
q1	q1	q0		✓

State diagram (minimal DFA)



Question 15

Language: Even number of both 0s and 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \} \quad (|Q| = 4)$

$\Sigma = \{ 0, 1 \}$

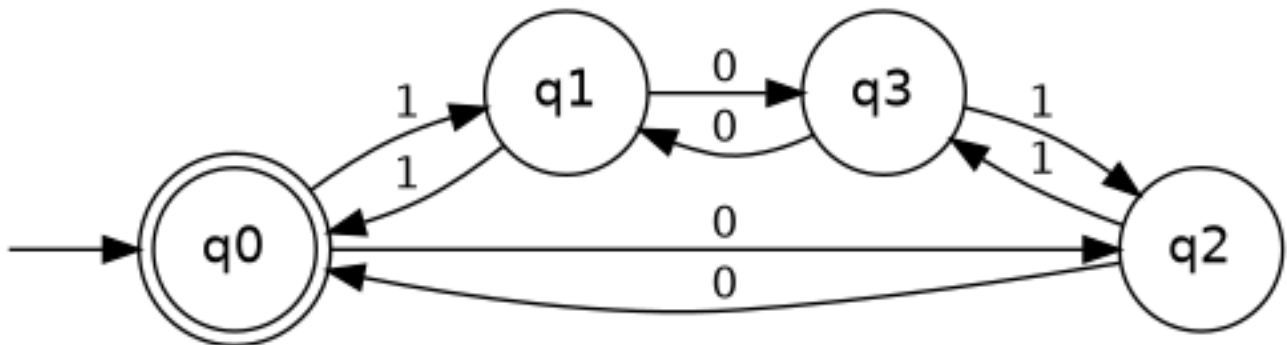
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q1	→	✓
q1	q3	q0		
q2	q0	q3		
q3	q1	q2		

State diagram (minimal DFA)



Question 16

Language: Odd number of both 0s and 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

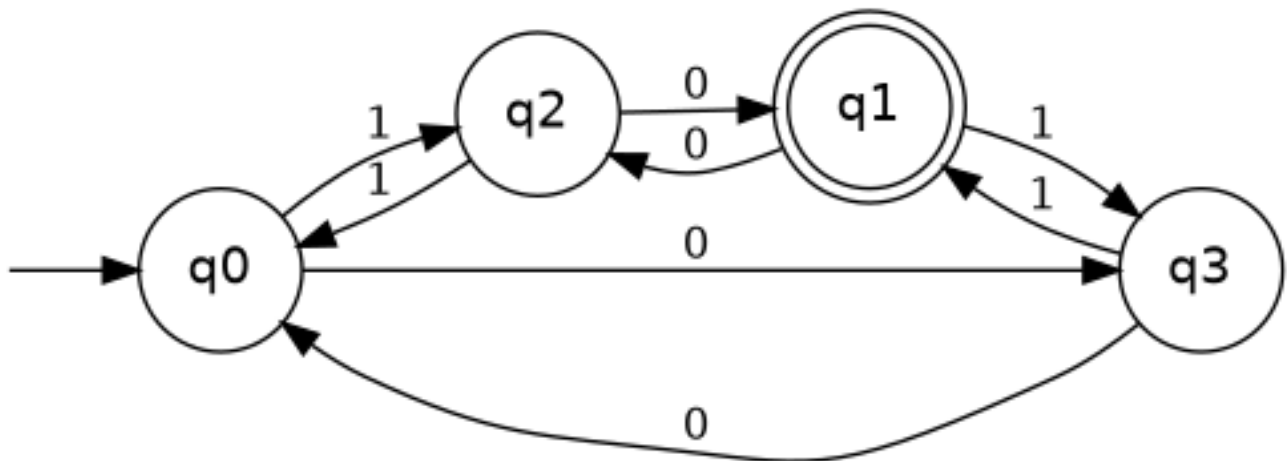
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q2	→	
q1	q2	q3		✓
q2	q1	q0		
q3	q0	q1		

State diagram (minimal DFA)



Question 17

Language: Even number of 0s and odd number of 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

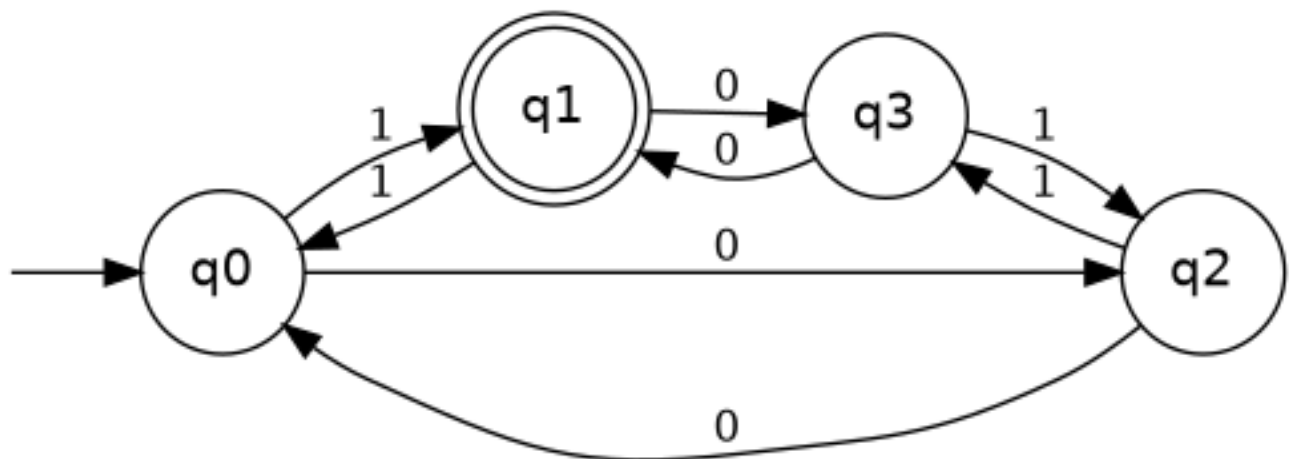
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q1	→	
q1	q3	q0		✓
q2	q0	q3		
q3	q1	q2		

State diagram (minimal DFA)



Question 18

Language: Odd number of 0s and even number of 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

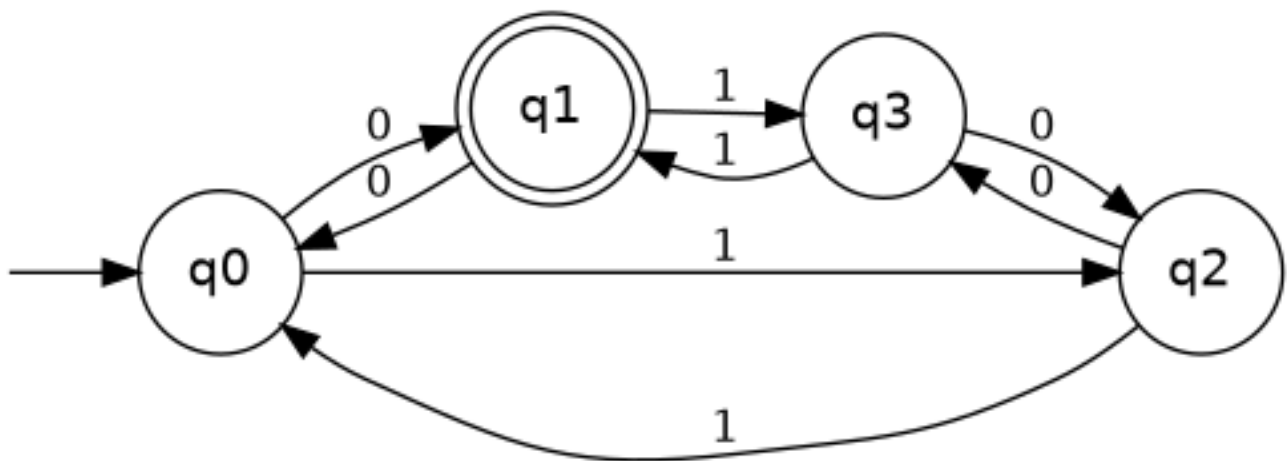
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q2	→	
q1	q0	q3		✓
q2	q3	q0		
q3	q2	q1		

State diagram (minimal DFA)



Question 19

Language: Exactly two 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

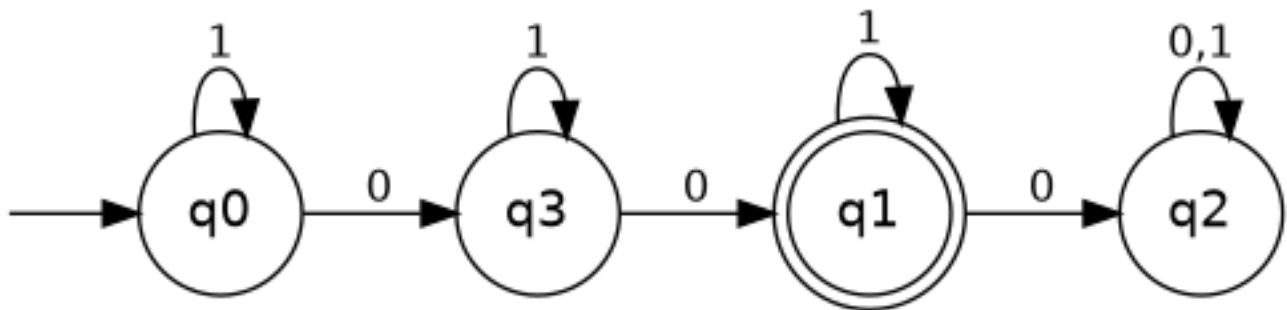
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q2	q1		✓
q2	q2	q2		
q3	q1	q3		

State diagram (minimal DFA)



Question 20

Language: Exactly three 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

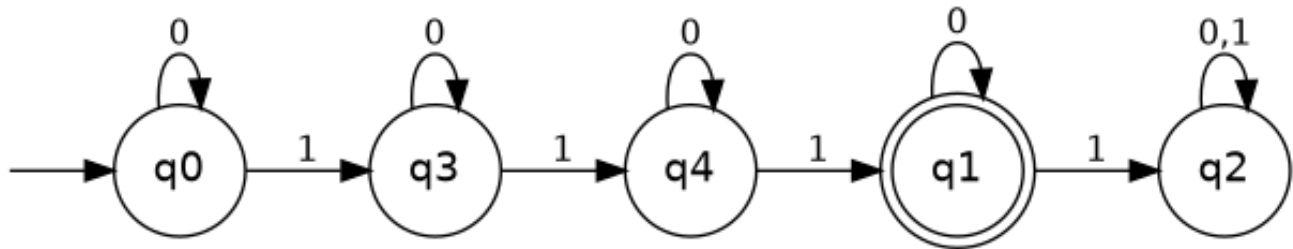
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q3	→	
q1	q1	q2		✓
q2	q2	q2		
q3	q3	q4		
q4	q4	q1		

State diagram (minimal DFA)



Question 21

Language: At least two 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

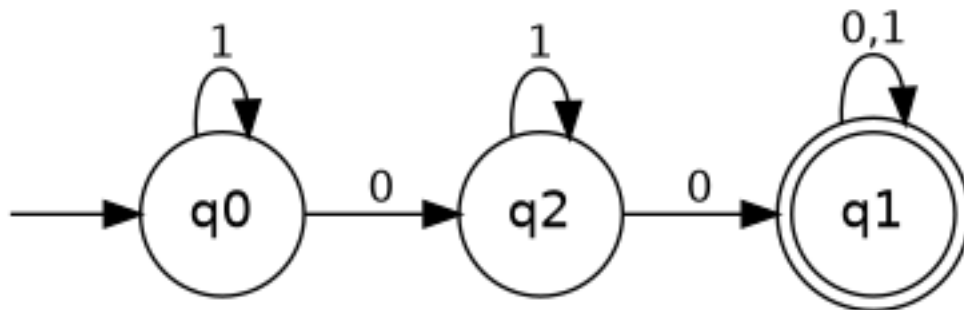
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q_0	q_2	q_0	$\rightarrow$	
q_1	q_1	q_1		✓
q_2	q_1	q_2		

State diagram (minimal DFA)



Question 22

Language: At most two 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \} \quad (|Q| = 4)$

$\Sigma = \{ 0, 1 \}$

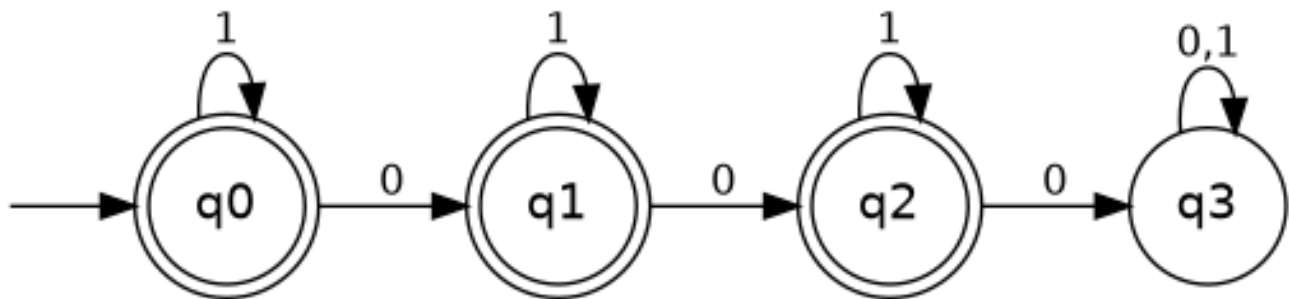
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q1		✓
q2	q3	q2		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 23

Language: At least three 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

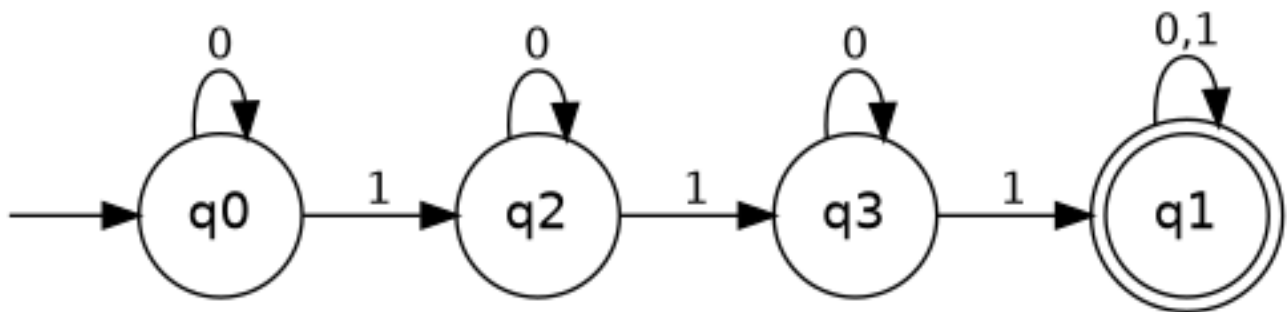
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q2	q3		
q3	q3	q1		

State diagram (minimal DFA)



Question 24

Language: At most three 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

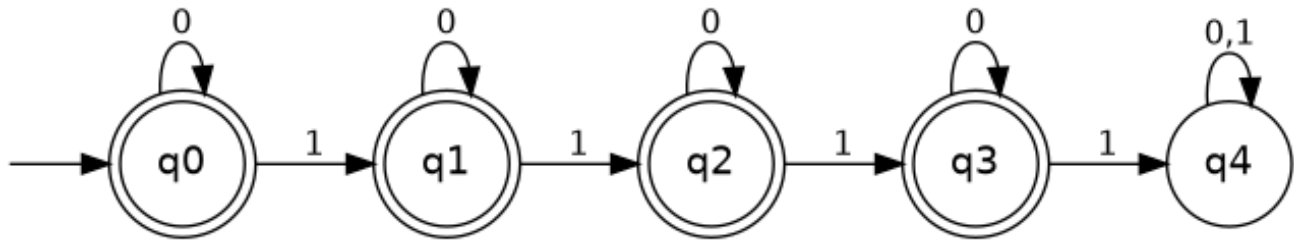
$q_0 = q_0$

$F = \{ q_0, q_1, q_2, q_3 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q1	q2		✓
q2	q2	q3		✓
q3	q3	q4		✓
q4	q4	q4		

State diagram (minimal DFA)



Question 25

Language: Exactly two 0s and one 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6 \} \quad (|Q| = 7)$

$\Sigma = \{ 0, 1 \}$

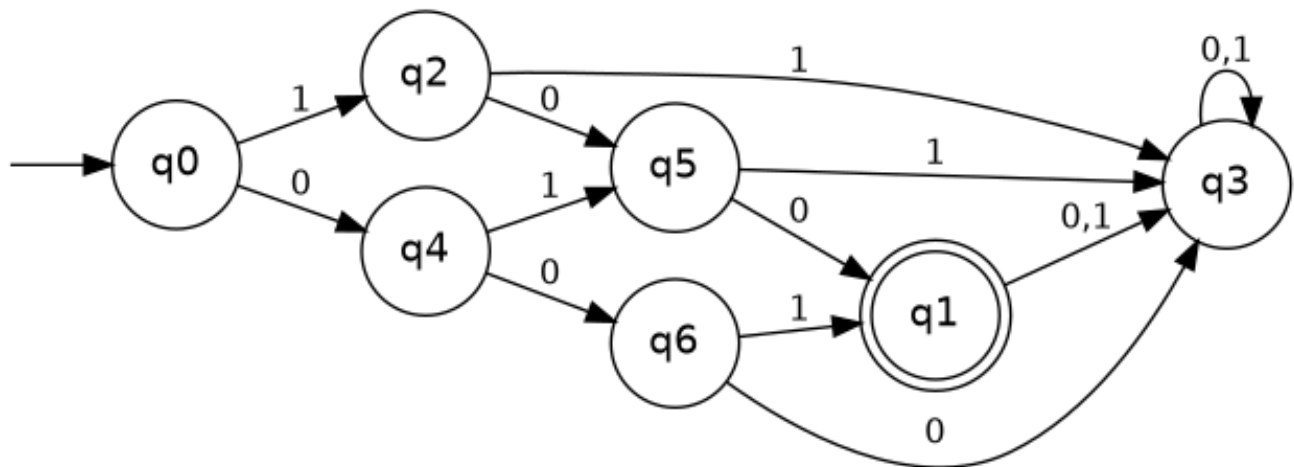
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q4	q2	→	
q1	q3	q3		✓
q2	q5	q3		
q3	q3	q3		
q4	q6	q5		
q5	q1	q3		
q6	q3	q1		

State diagram (minimal DFA)



Question 26

Language: More than two 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

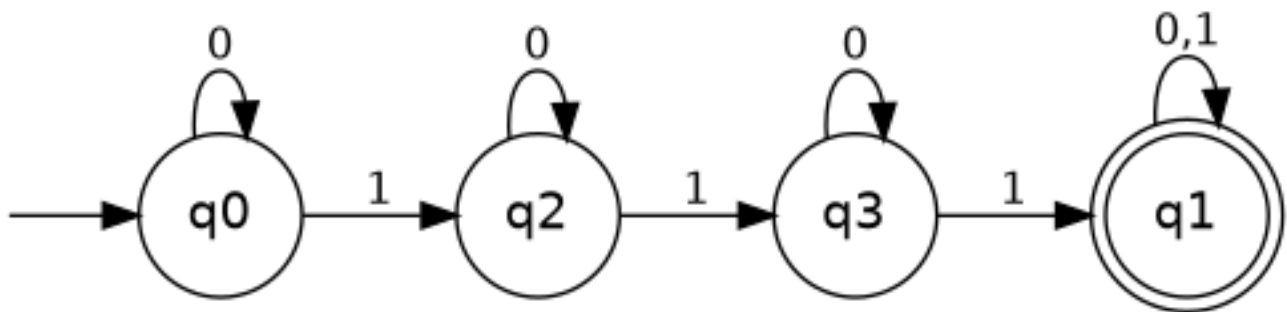
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q2	q3		
q3	q3	q1		

State diagram (minimal DFA)



Question 27

Language: Fewer than three 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

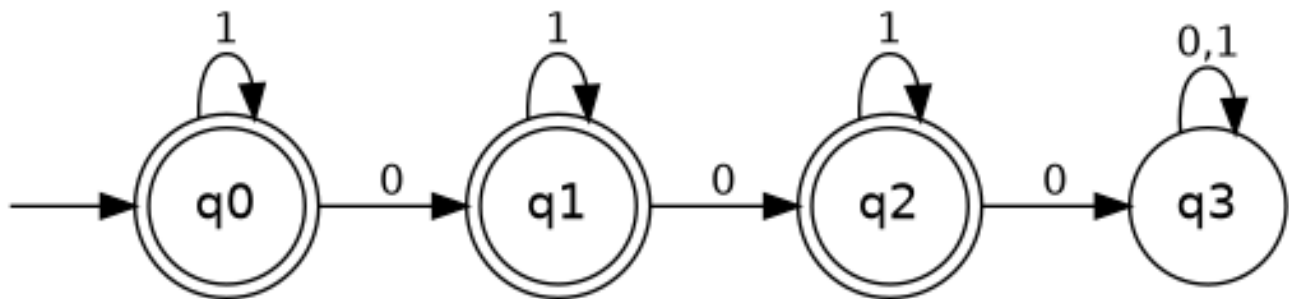
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q1		✓
q2	q3	q2		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 28

Language: Exactly four 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

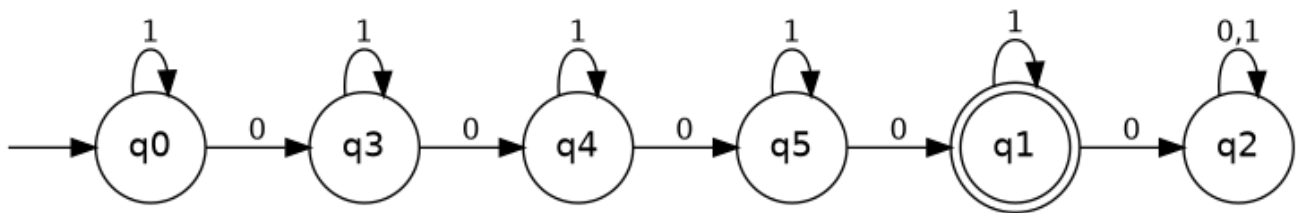
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q2	q1		✓
q2	q2	q2		
q3	q4	q3		
q4	q5	q4		
q5	q1	q5		

State diagram (minimal DFA)



Question 29

Language: Ending with 01.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

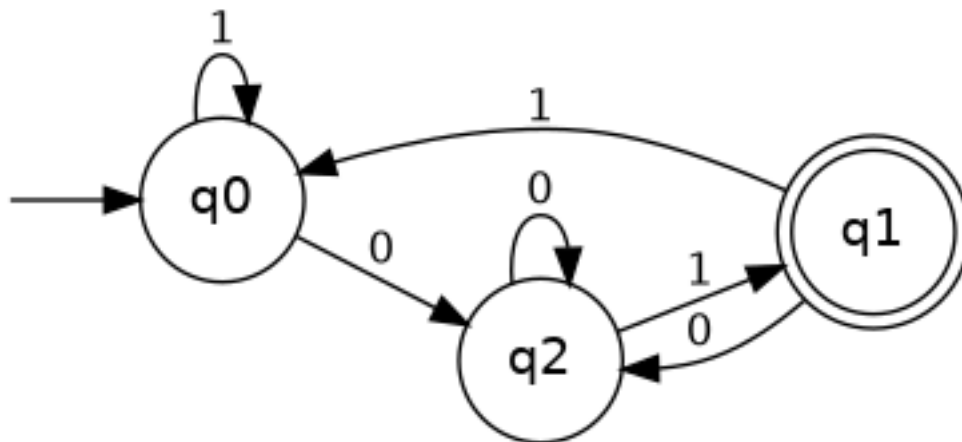
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q2	q0		✓
q2	q2	q1		

State diagram (minimal DFA)



Question 30

Language: Ending with 10.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

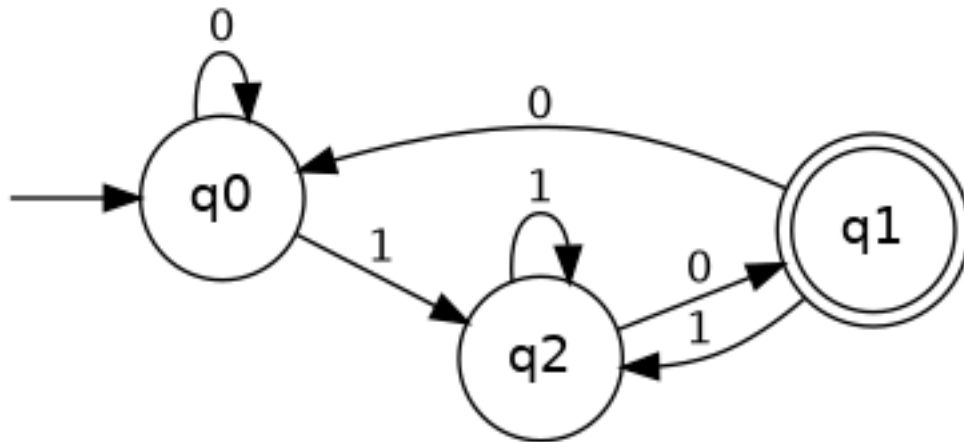
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q0	q2		✓
q2	q1	q2		

State diagram (minimal DFA)



Question 31

Language: Ending with 00.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

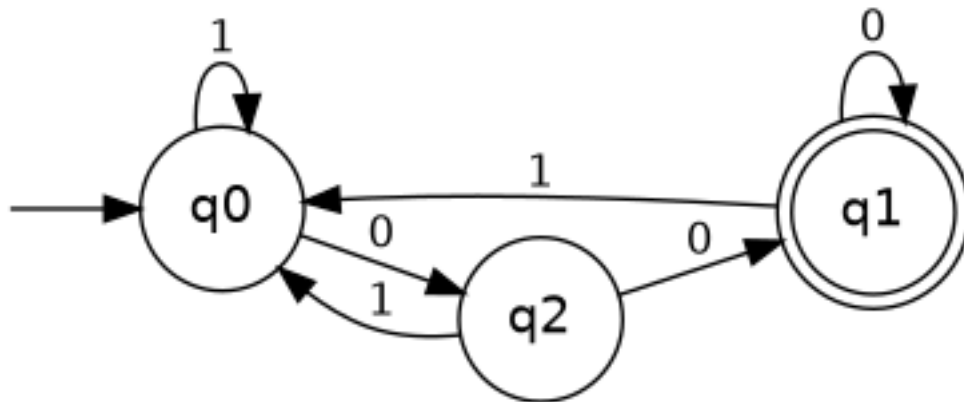
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q1	q0		✓
q2	q1	q0		

State diagram (minimal DFA)



Question 32

Language: Ending with 11.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

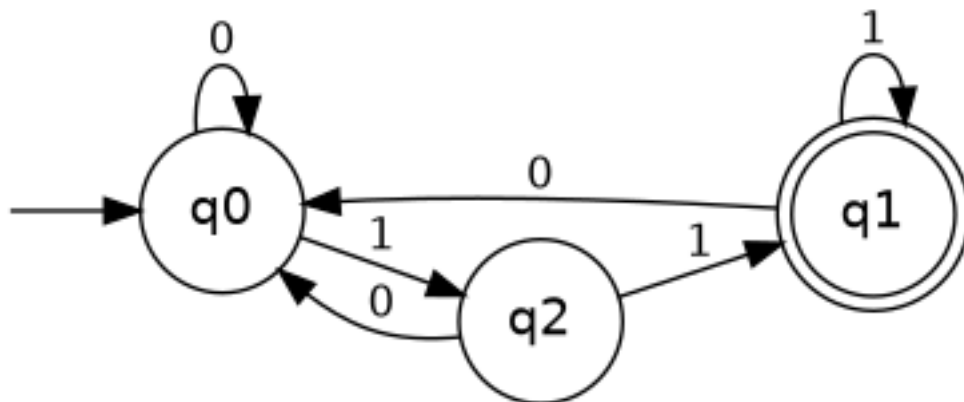
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q0	q1		✓
q2	q0	q1		

State diagram (minimal DFA)



Question 33

Language: Ending with 001.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

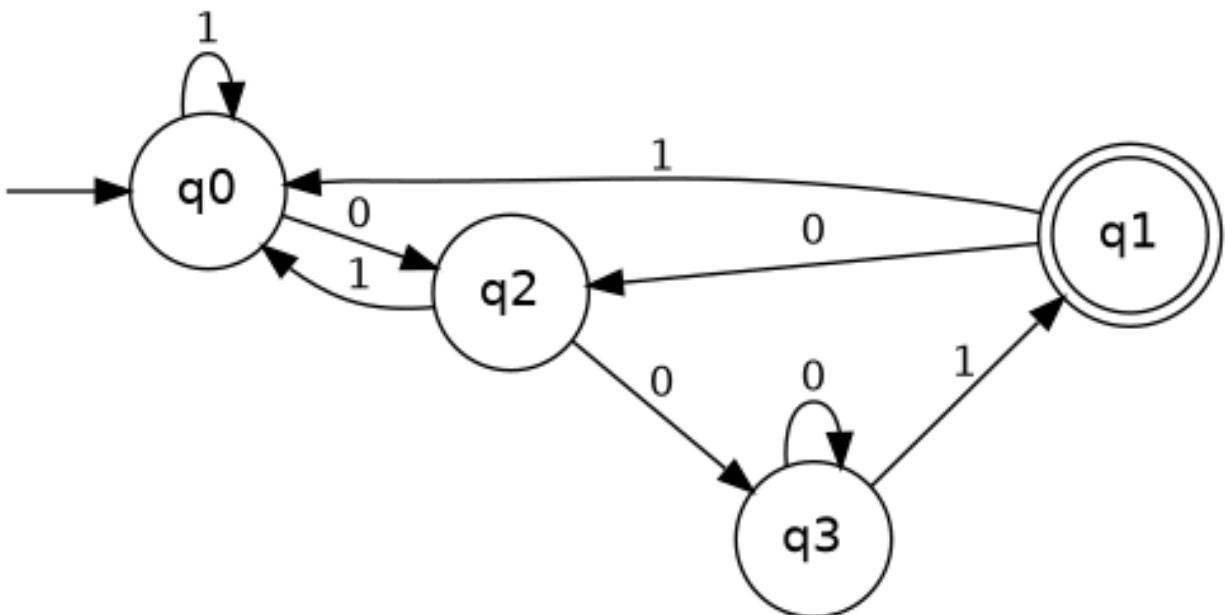
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q2	q0		✓
q2	q3	q0		
q3	q3	q1		

State diagram (minimal DFA)



Question 34

Language: Ending with 101.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

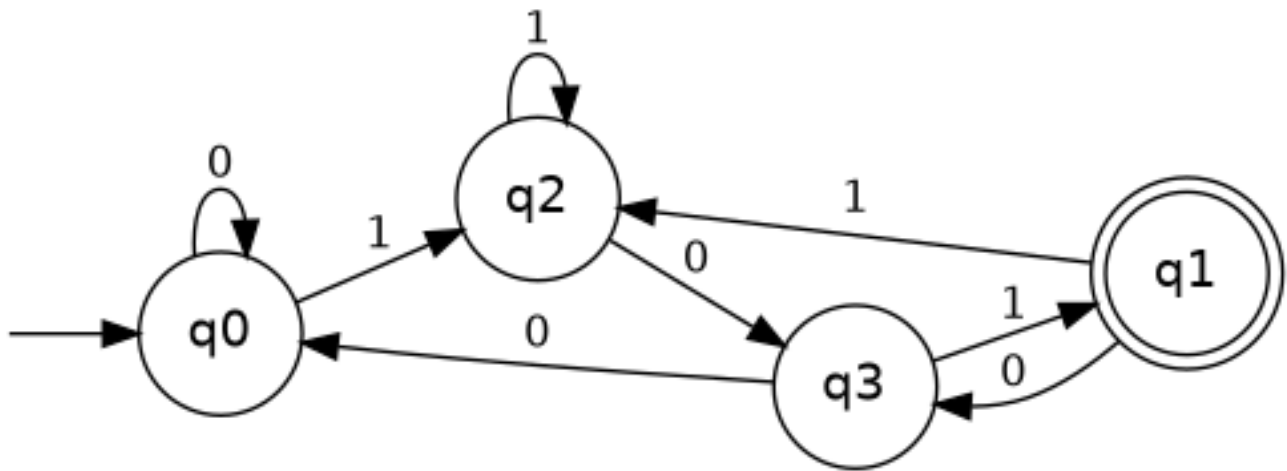
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q3	q2		✓
q2	q3	q2		
q3	q0	q1		

State diagram (minimal DFA)



Question 35

Language: Starting with 01.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

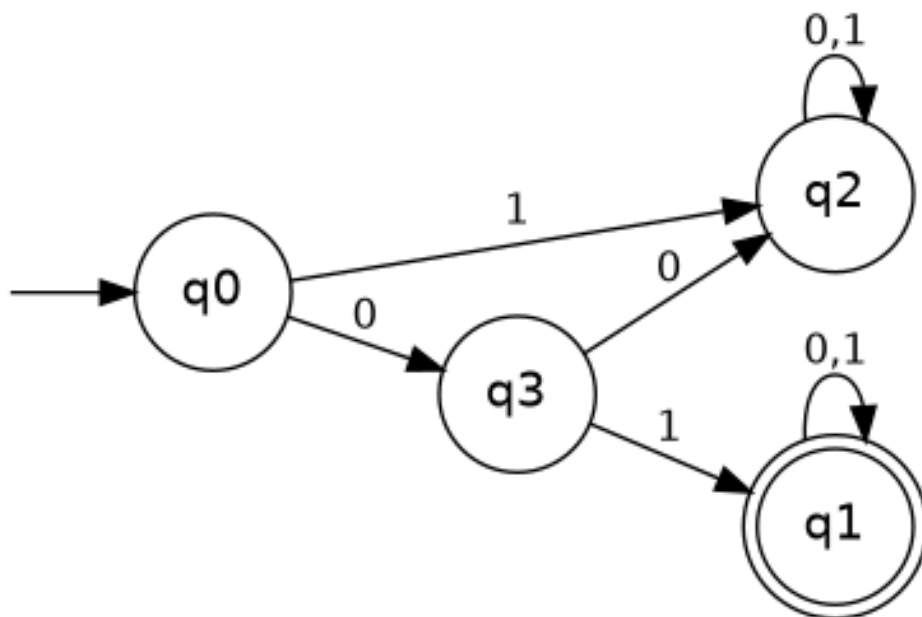
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q2	→	
q1	q1	q1		✓
q2	q2	q2		
q3	q2	q1		

State diagram (minimal DFA)



Question 36

Language: Starting with 10.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

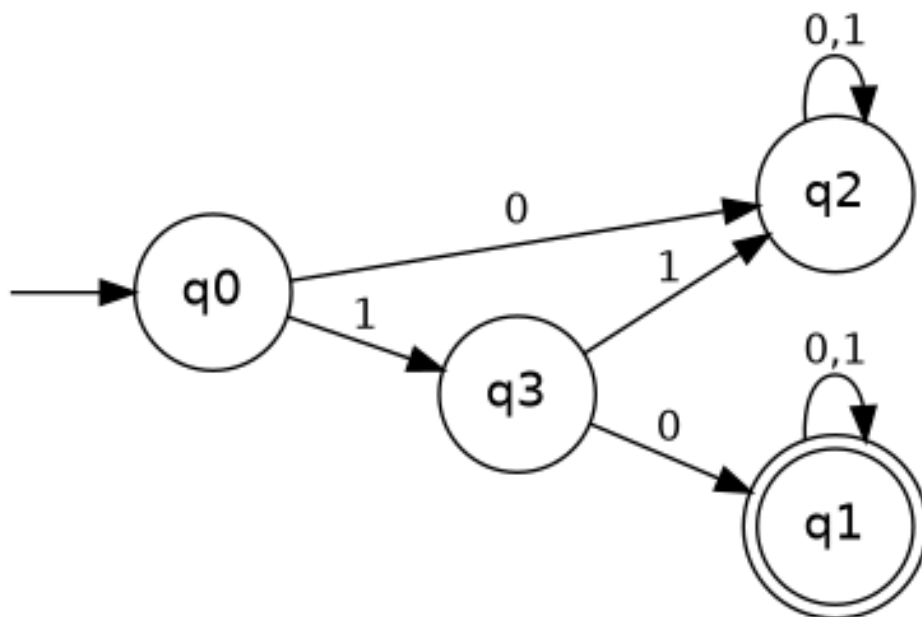
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q3	→	
q1	q1	q1		✓
q2	q2	q2		
q3	q1	q2		

State diagram (minimal DFA)



Question 37

Language: Starting with 001.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

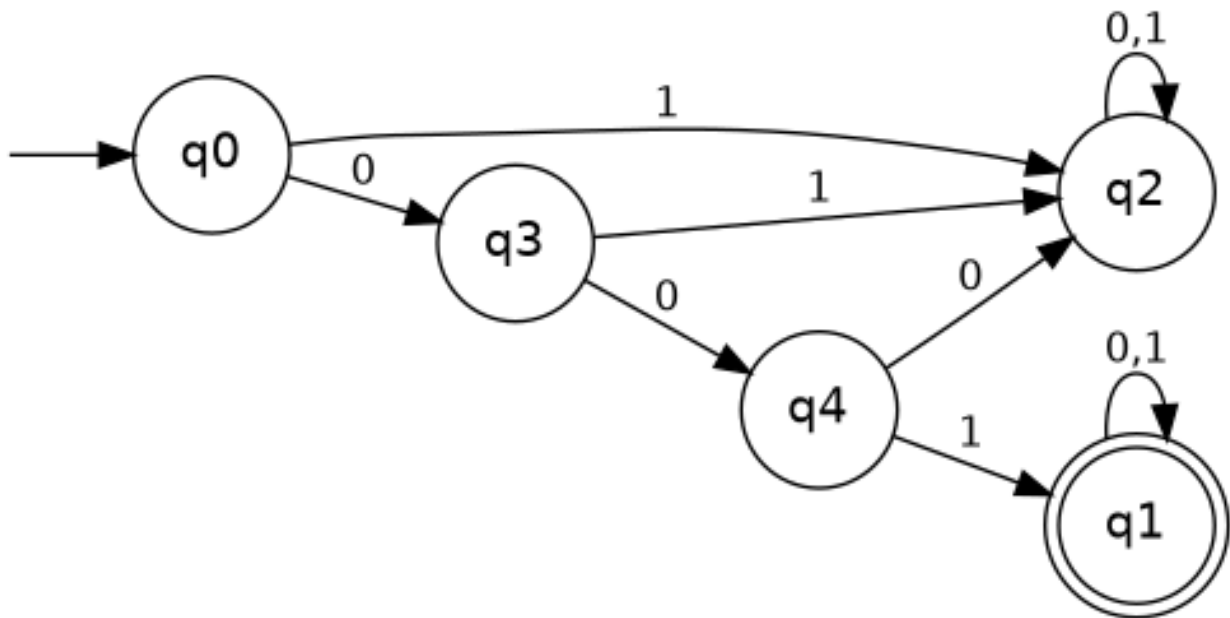
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q2	→	
q1	q1	q1		✓
q2	q2	q2		
q3	q4	q2		
q4	q2	q1		

State diagram (minimal DFA)



Question 38

Language: Starting with 111.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

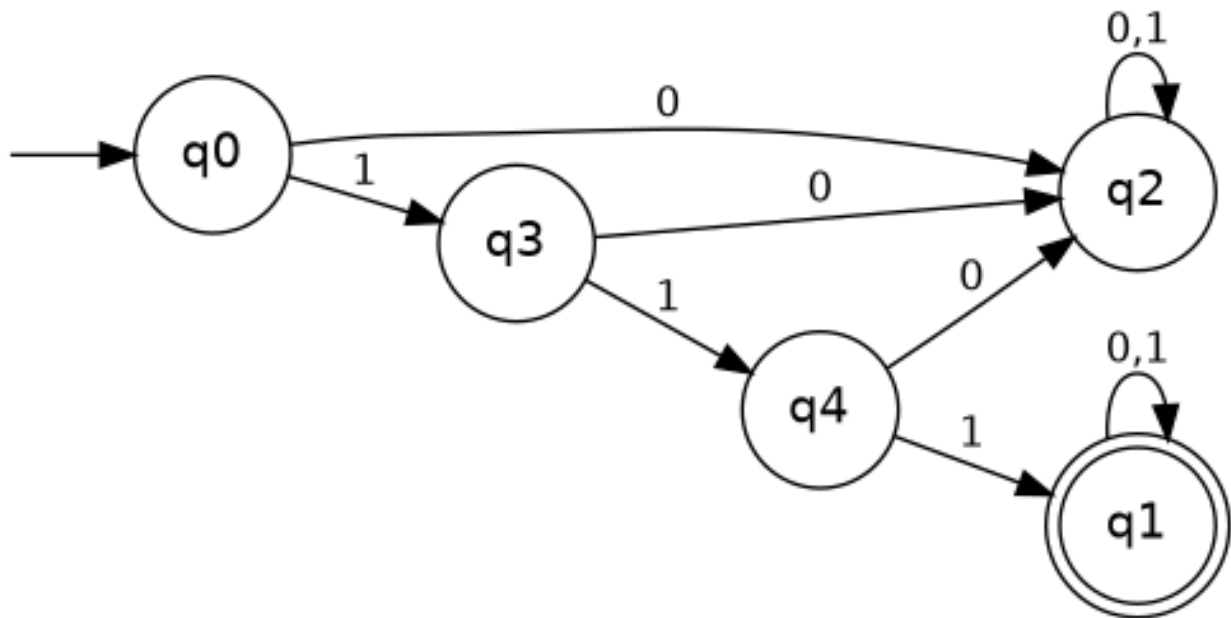
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q3	→	
q1	q1	q1		✓
q2	q2	q2		
q3	q2	q4		
q4	q2	q1		

State diagram (minimal DFA)



Question 39

Language: Containing substring 01.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

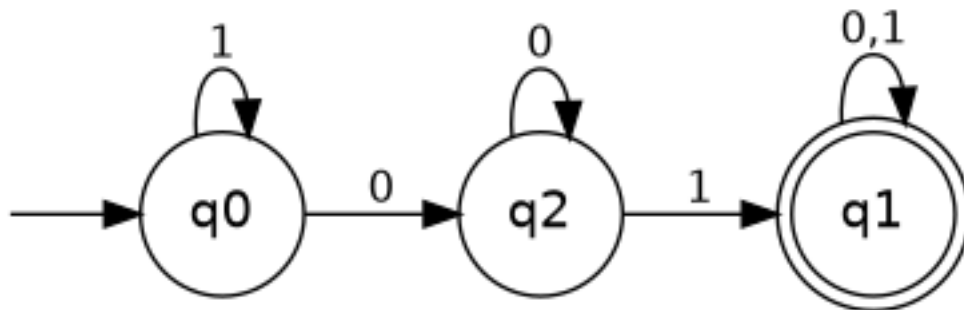
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q1	q1		✓
q2	q2	q1		

State diagram (minimal DFA)



Question 40

Language: Containing substring 10.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

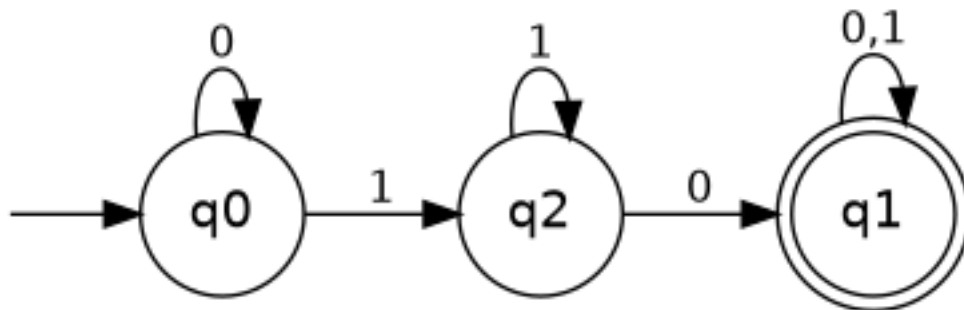
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q1	q2		

State diagram (minimal DFA)



Question 41

Language: Containing substring 101.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

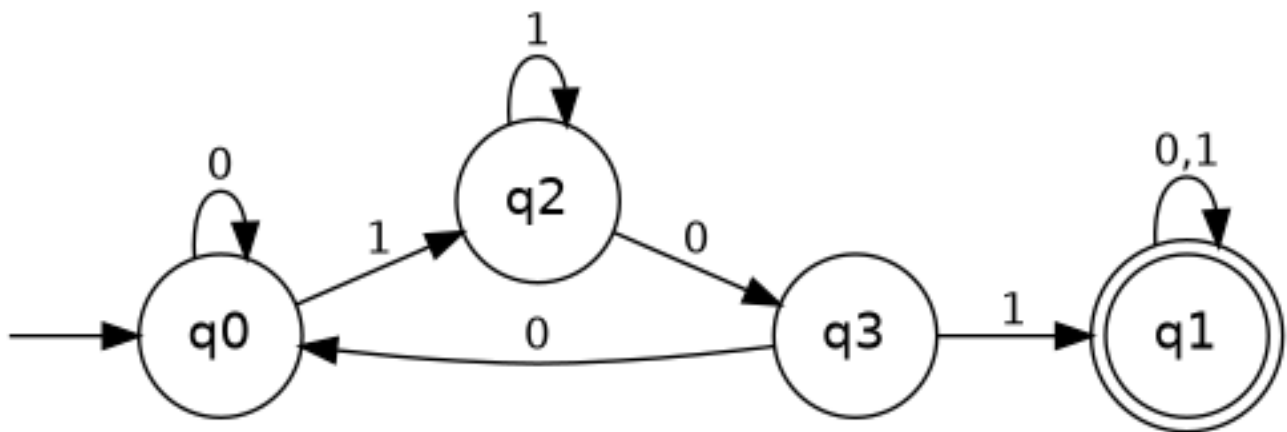
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q3	q2		
q3	q0	q1		

State diagram (minimal DFA)



Question 42

Language: Containing substring 110.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

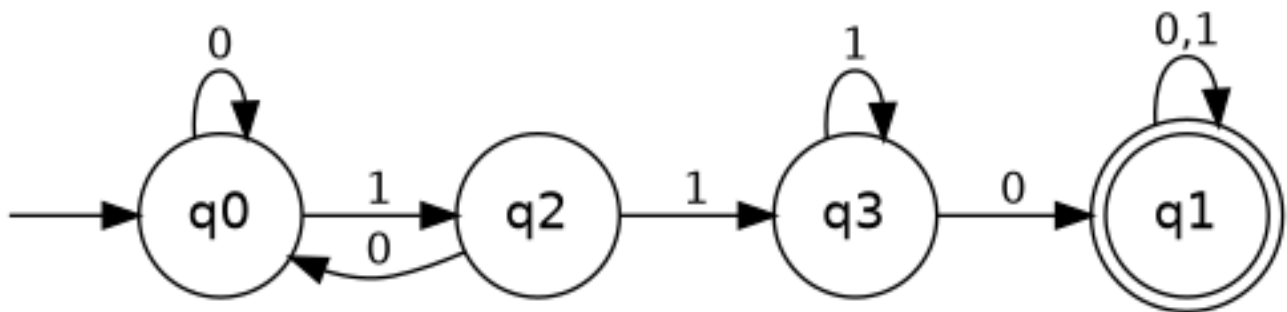
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q0	q3		
q3	q1	q3		

State diagram (minimal DFA)



Question 43

Language: Containing substring 001.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

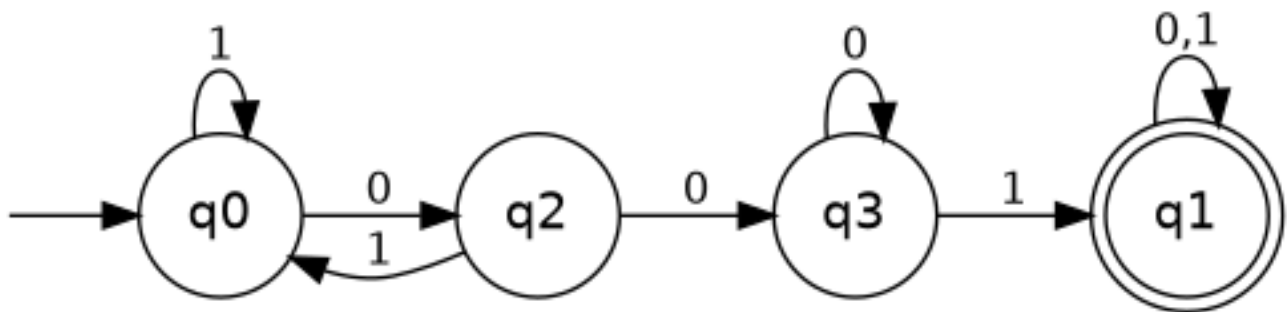
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q1	q1		✓
q2	q3	q0		
q3	q3	q1		

State diagram (minimal DFA)



Question 44

Language: Containing substring 111.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

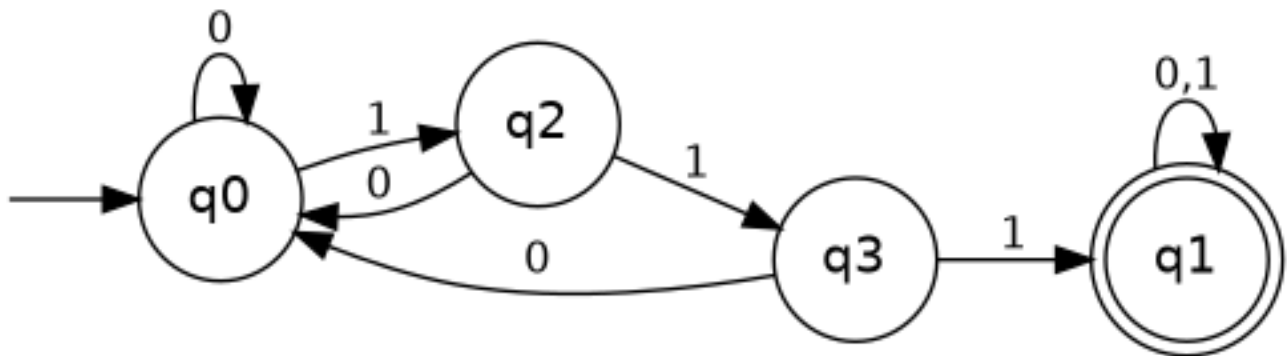
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q1		✓
q2	q0	q3		
q3	q0	q1		

State diagram (minimal DFA)



Question 45

Language: Containing substring 000.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

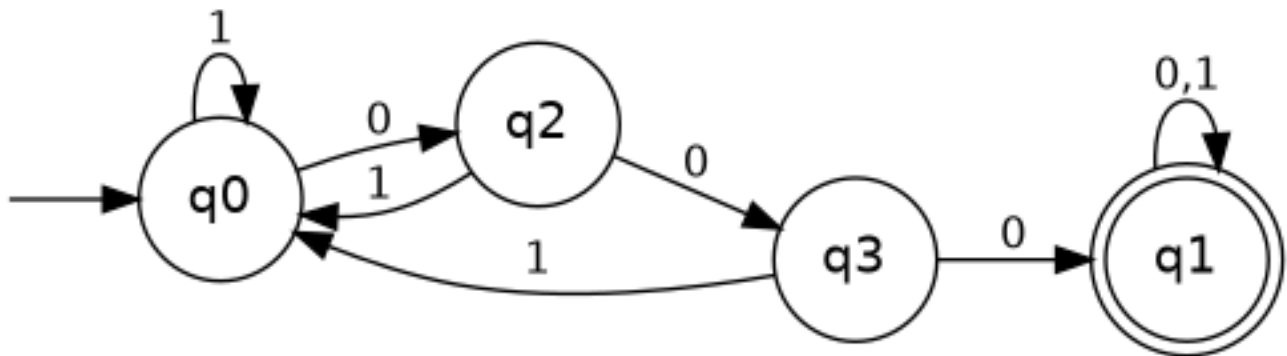
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q1	q1		✓
q2	q3	q0		
q3	q1	q0		

State diagram (minimal DFA)



Question 46

Language: Containing substring 0101.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

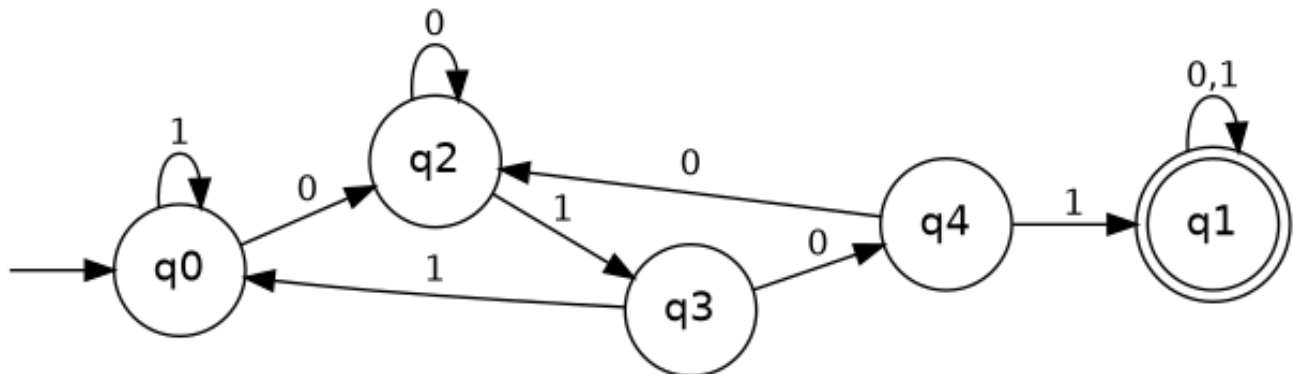
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q1	q1		✓
q2	q2	q3		
q3	q4	q0		
q4	q2	q1		

State diagram (minimal DFA)



Question 47

Language: Do not contain 11.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

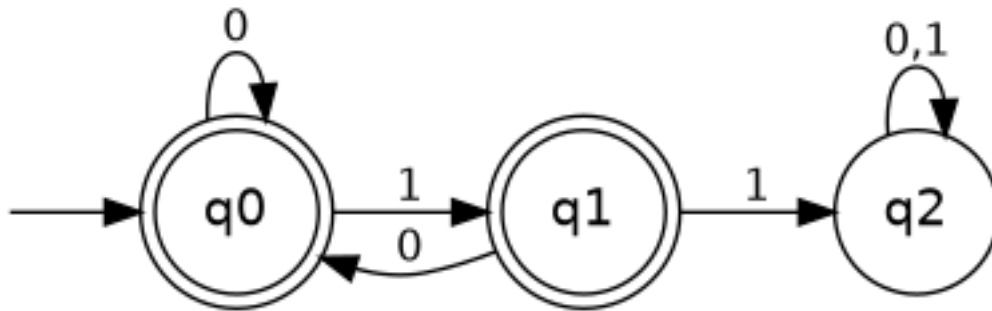
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q0	q2		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 48

Language: Do not contain 00.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

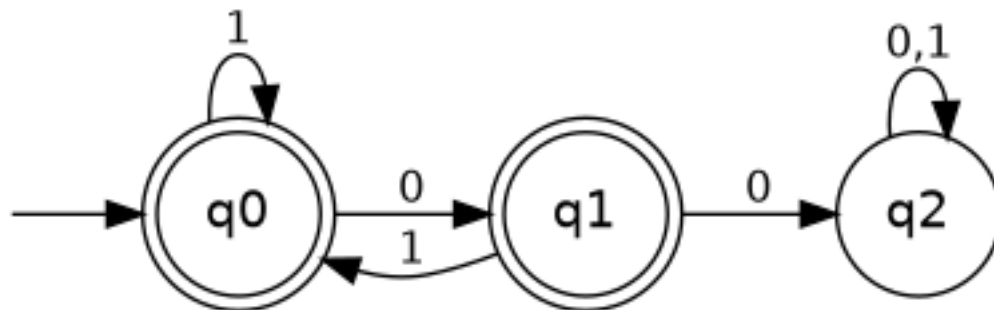
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 49

Language: Do not contain 101.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

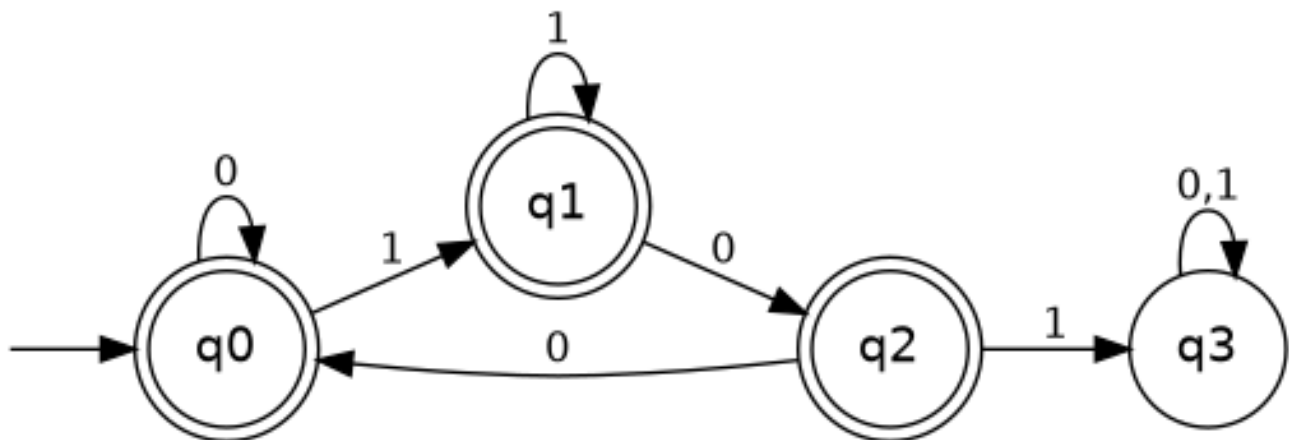
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q2	q1		✓
q2	q0	q3		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 50

Language: Do not contain 010.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

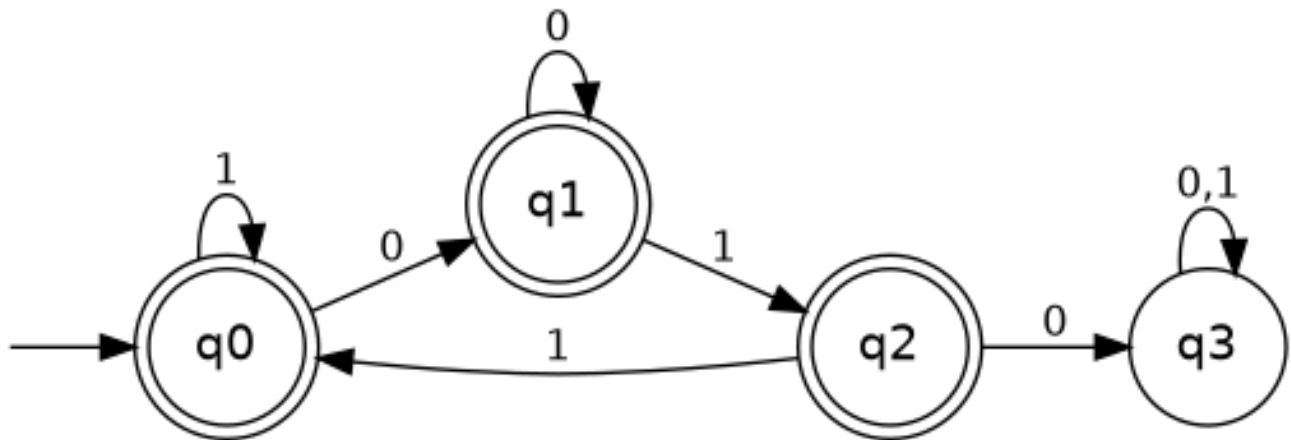
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q1	q2		✓
q2	q3	q0		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 51

Language: No consecutive 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

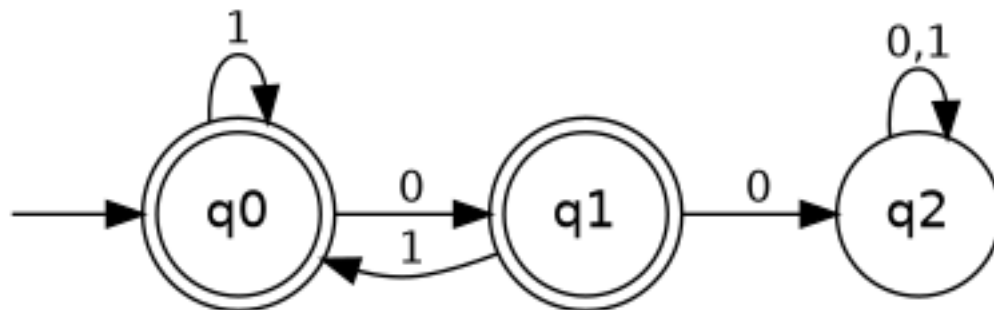
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 52

Language: No consecutive 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

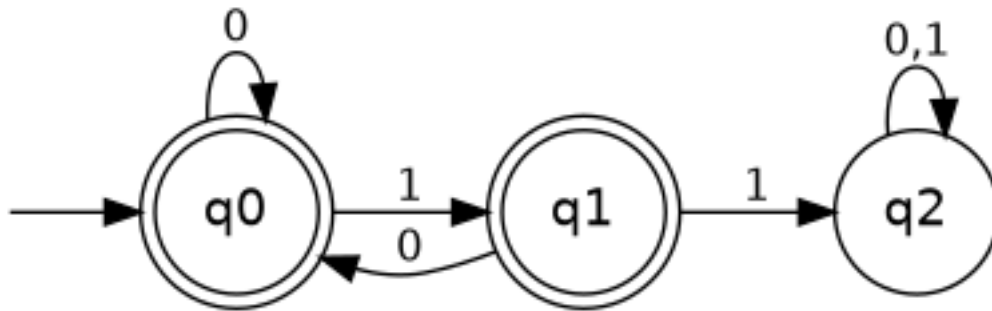
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q0	q2		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 53

Language: Do not contain 111.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

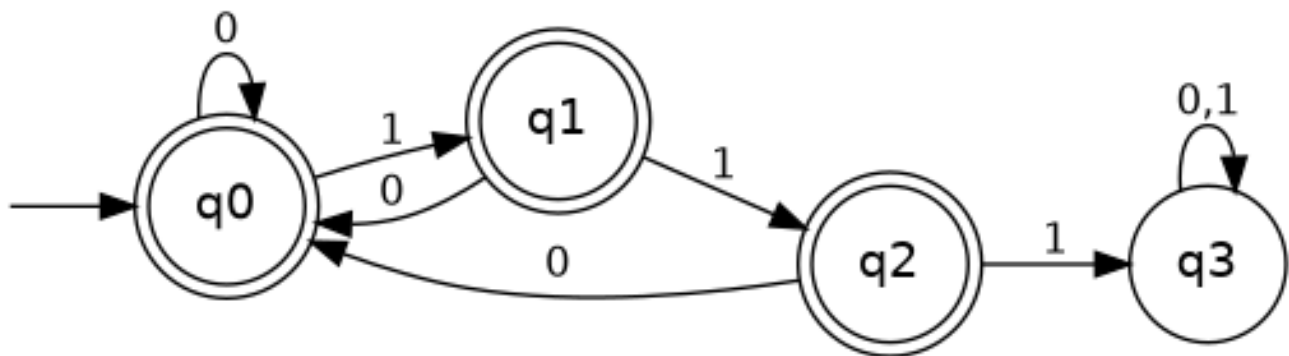
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q0	q2		✓
q2	q0	q3		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 54

Language: Do not contain 001.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

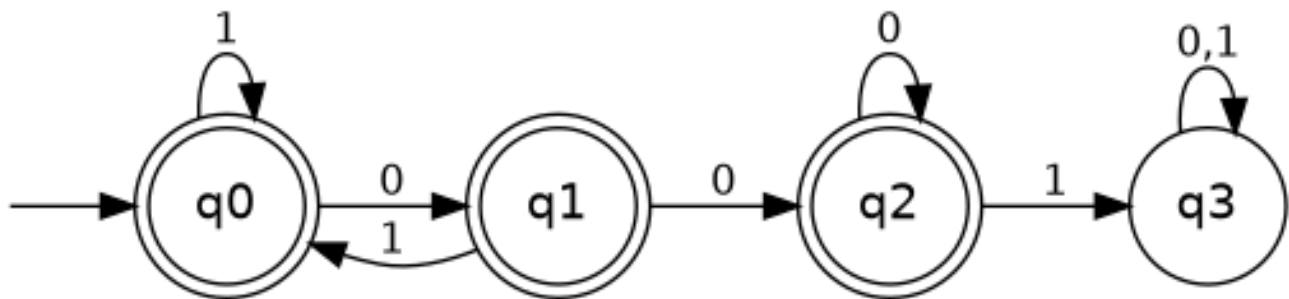
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		✓
q2	q2	q3		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 55

Language: Even length.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q1	→	✓
q1	q0	q0		

State diagram (minimal DFA)



Question 56

Language: Odd length.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

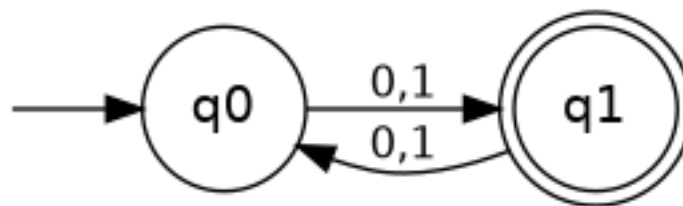
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q1	→	
q1	q0	q0		✓

State diagram (minimal DFA)



Question 57

Language: Length divisible by 3.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

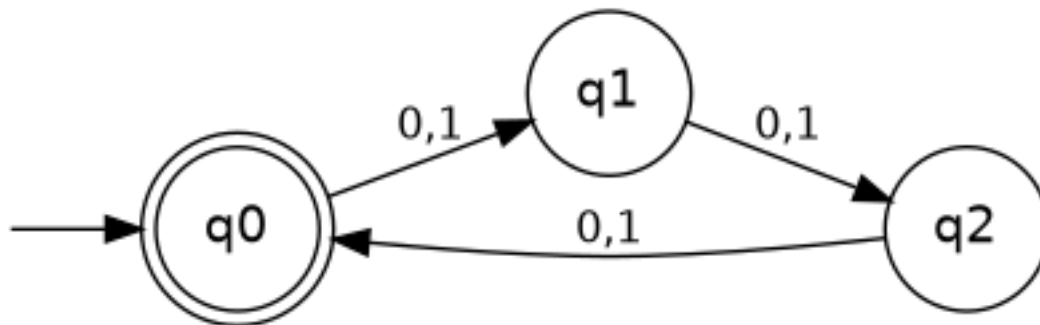
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q1	→	✓
q1	q2	q2		
q2	q0	q0		

State diagram (minimal DFA)



Question 58

Language: Length divisible by 4.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

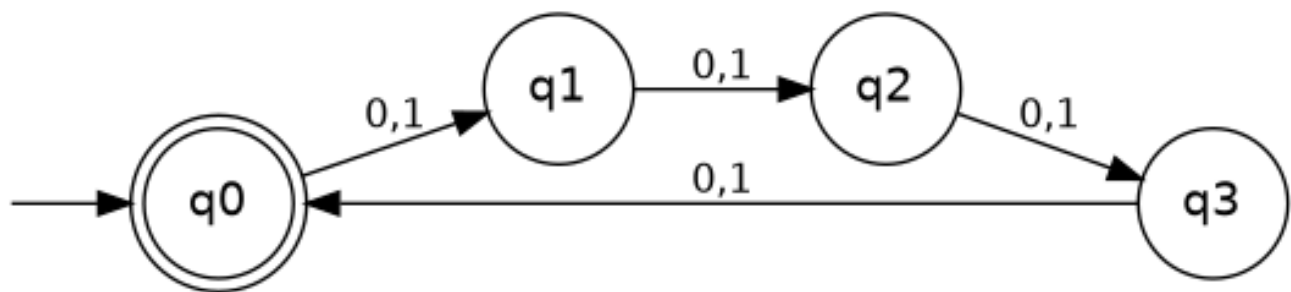
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q1	→	✓
q1	q2	q2		
q2	q3	q3		
q3	q0	q0		

State diagram (minimal DFA)



Question 59

Language: Length greater than 3.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

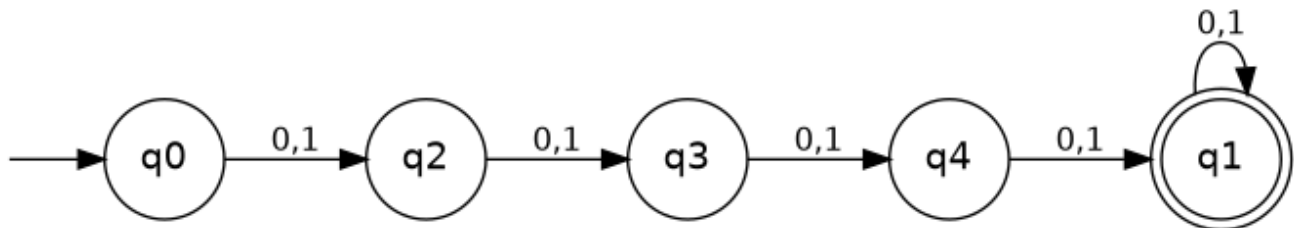
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q2	→	
q1	q1	q1		✓
q2	q3	q3		
q3	q4	q4		
q4	q1	q1		

State diagram (minimal DFA)



Question 60

Language: Length ≤ 5 .

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6 \} \quad (|Q| = 7)$

$\Sigma = \{ 0, 1 \}$

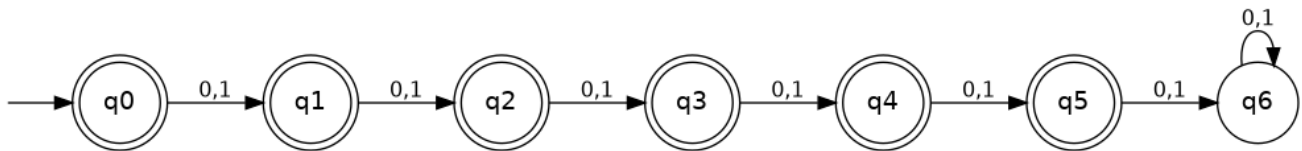
$q_0 = q_0$

$F = \{ q_0, q_1, q_2, q_3, q_4, q_5 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q1	→	✓
q1	q2	q2		✓
q2	q3	q3		✓
q3	q4	q4		✓
q4	q5	q5		✓
q5	q6	q6		✓
q6	q6	q6		

State diagram (minimal DFA)



Question 61

Language: Length exactly 4.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

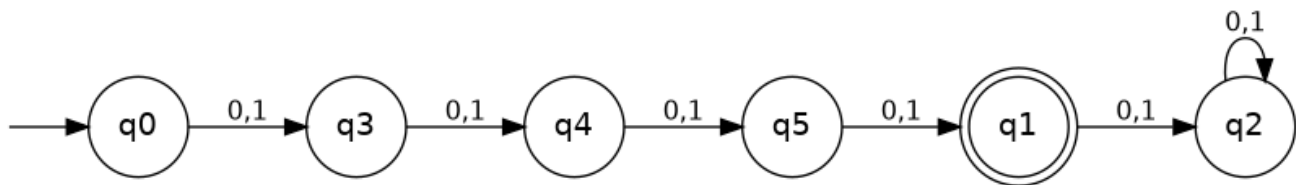
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q3	→	
q1	q2	q2		✓
q2	q2	q2		
q3	q4	q4		
q4	q5	q5		
q5	q1	q1		

State diagram (minimal DFA)



Question 62

Language: Length at least 5.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

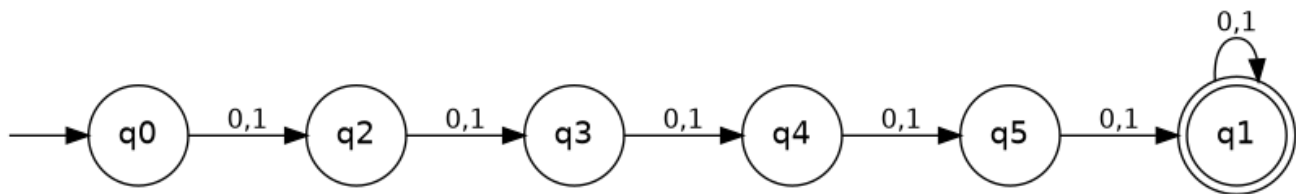
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q2	→	
q1	q1	q1		✓
q2	q3	q3		
q3	q4	q4		
q4	q5	q5		
q5	q1	q1		

State diagram (minimal DFA)



Question 63

Language: Binary numbers divisible by 2.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

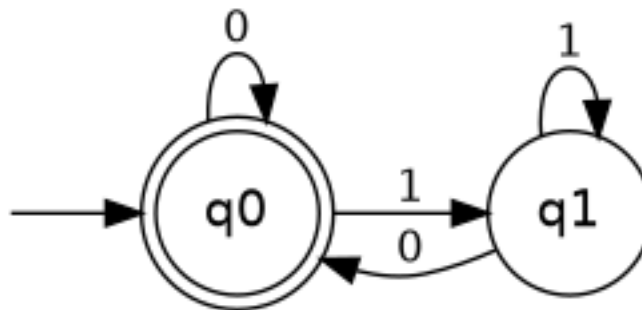
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q0	q1		

State diagram (minimal DFA)



Question 64

Language: Binary numbers divisible by 3.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

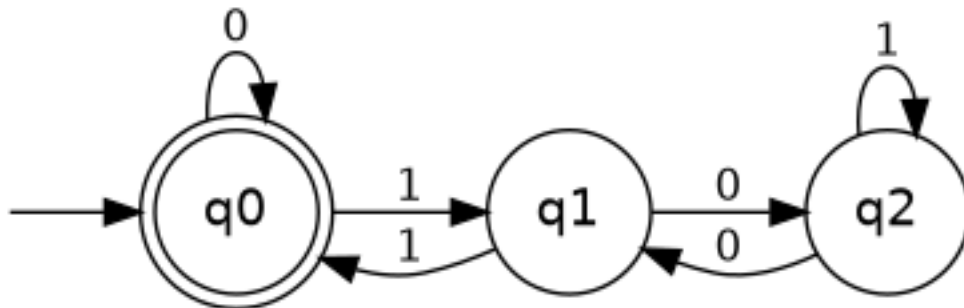
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q2	q0		
q2	q1	q2		

State diagram (minimal DFA)



Question 65

Language: Binary numbers divisible by 4.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

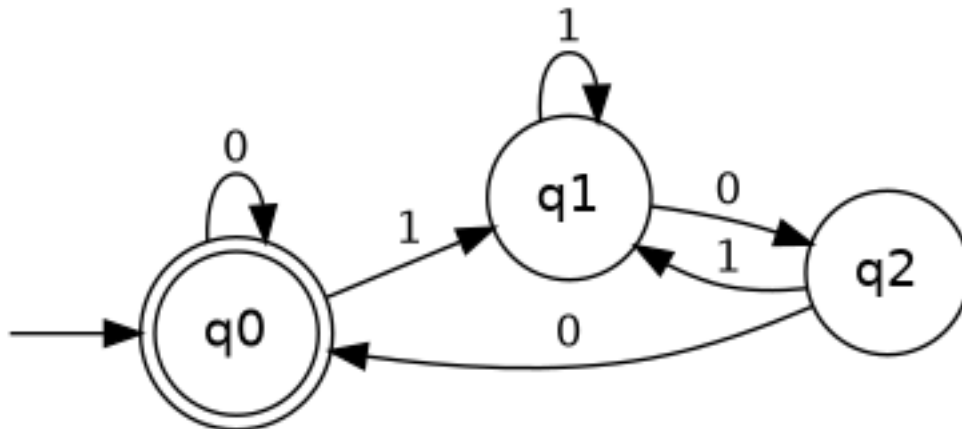
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q2	q1		
q2	q0	q1		

State diagram (minimal DFA)



Question 66

Language: Binary numbers divisible by 5.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

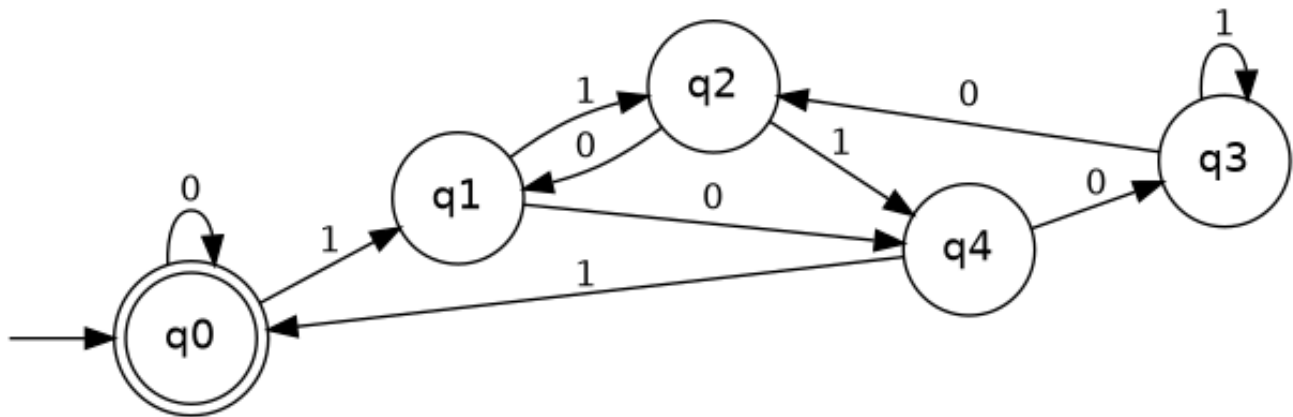
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q4	q2		
q2	q1	q4		
q3	q2	q3		
q4	q3	q0		

State diagram (minimal DFA)



Question 67

Language: Binary numbers divisible by 6.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

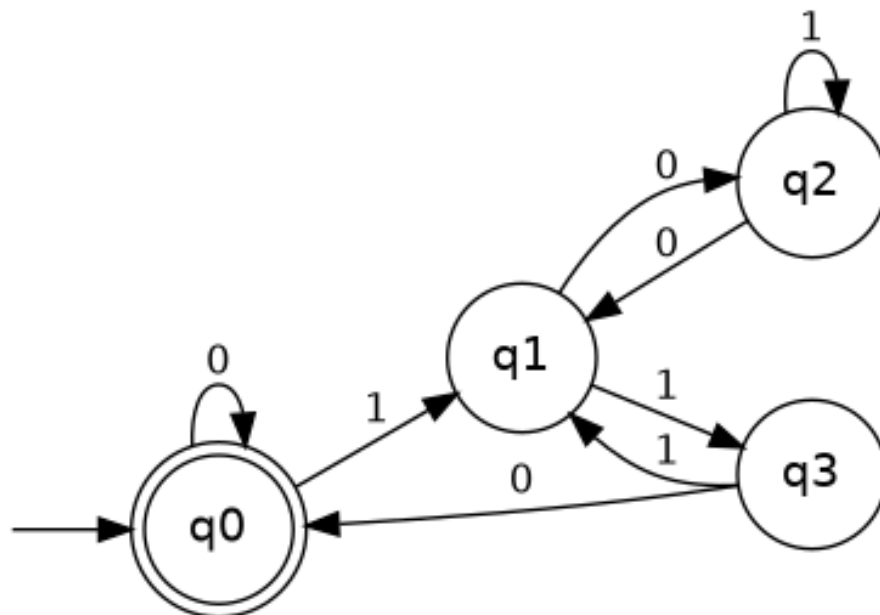
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q2	q3		
q2	q1	q2		
q3	q0	q1		

State diagram (minimal DFA)



Question 68

Language: Binary numbers divisible by 7.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6 \} \quad (|Q| = 7)$

$\Sigma = \{ 0, 1 \}$

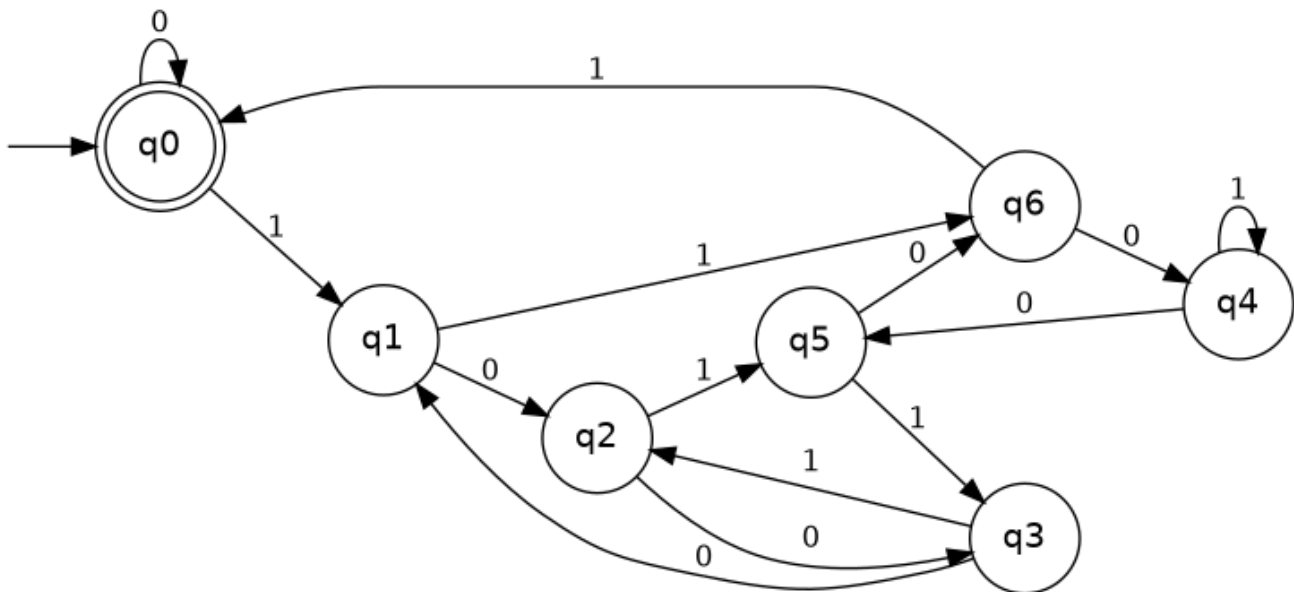
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q2	q6		
q2	q3	q5		
q3	q1	q2		
q4	q5	q4		
q5	q6	q3		
q6	q4	q0		

State diagram (minimal DFA)



Question 69

Language: Ending with 01 and even number of 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

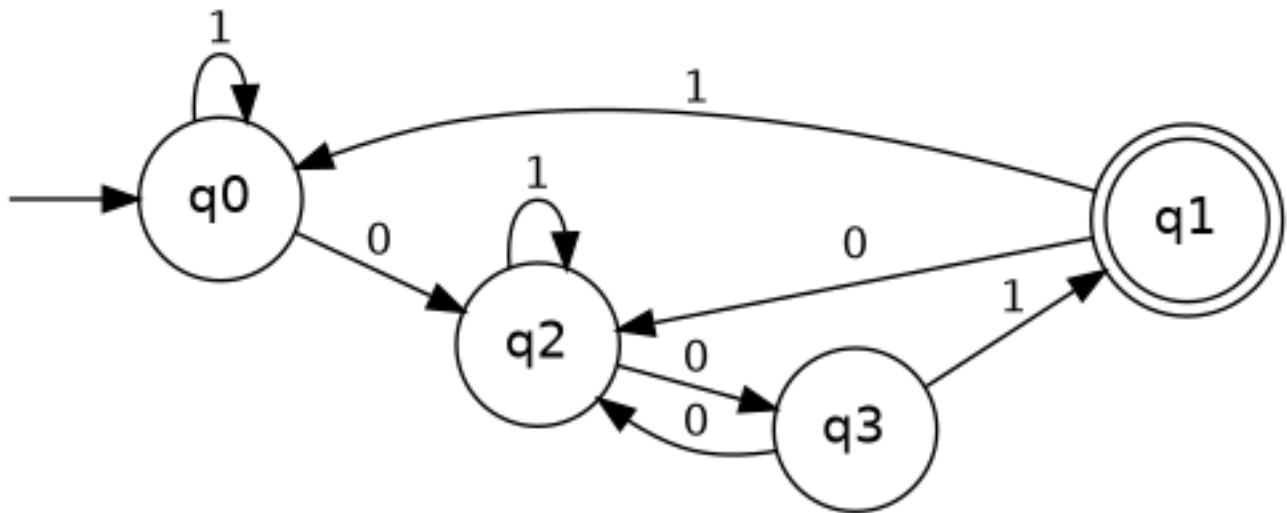
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q2	q0		✓
q2	q3	q2		
q3	q2	q1		

State diagram (minimal DFA)



Question 70

Language: Starting with 1 and ending with 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

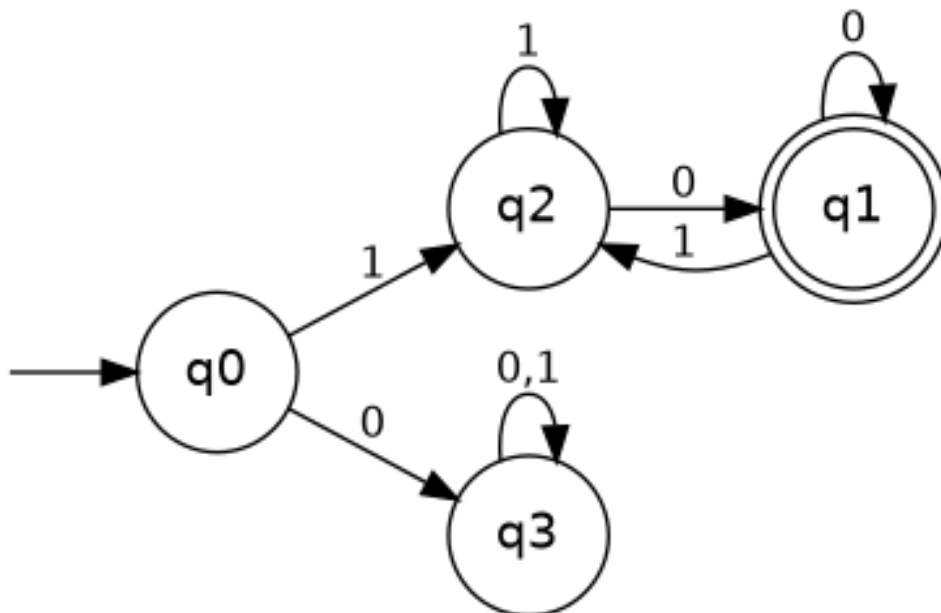
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q2	→	
q1	q1	q2		✓
q2	q1	q2		
q3	q3	q3		

State diagram (minimal DFA)



Question 71

Language: Even number of 1s and contains 00.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \}$ ($|Q| = 6$)

$\Sigma = \{ 0, 1 \}$

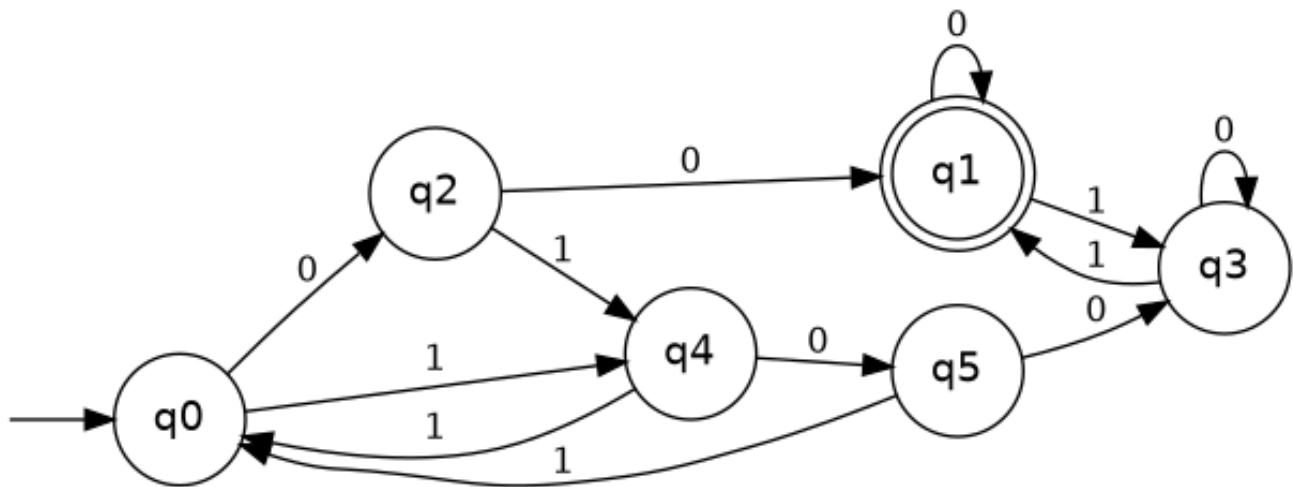
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q4	→	
q1	q1	q3		✓
q2	q1	q4		
q3	q3	q1		
q4	q5	q0		
q5	q3	q0		

State diagram (minimal DFA)



Question 72

Language: Starting with 01 and exactly two 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

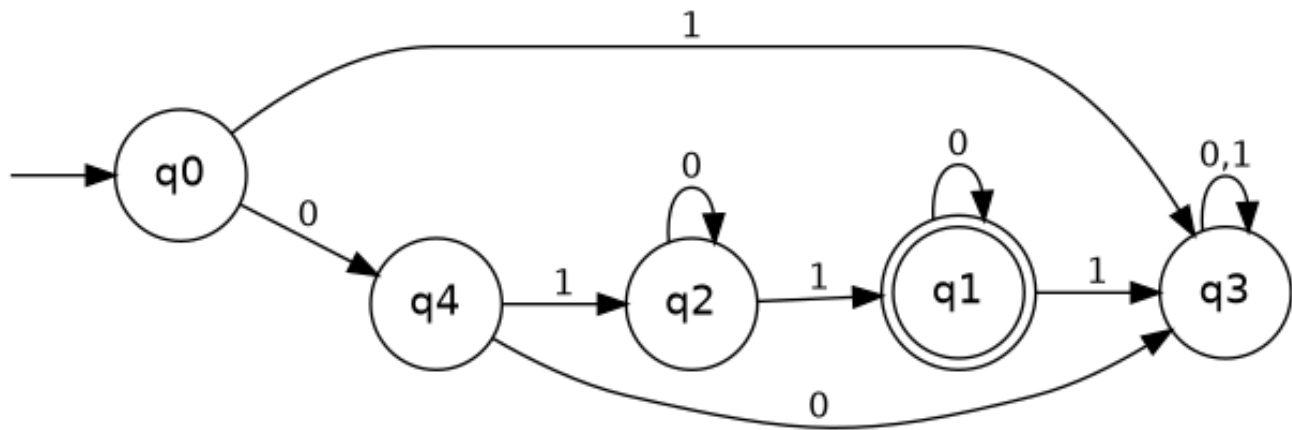
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q4	q3	→	
q1	q1	q3		✓
q2	q2	q1		
q3	q3	q3		
q4	q3	q2		

State diagram (minimal DFA)



Question 73

Language: Ending with 10 and odd number of 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

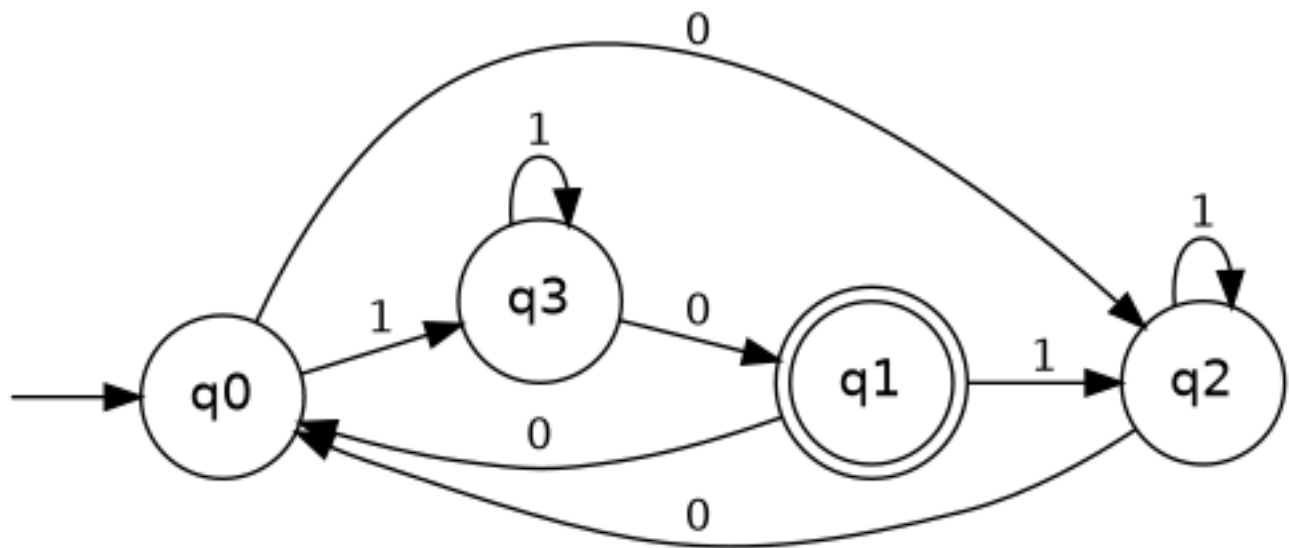
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q3	→	
q1	q0	q2		✓
q2	q0	q2		
q3	q1	q3		

State diagram (minimal DFA)



Question 74

Language: Beginning with 1 and length divisible by 3.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

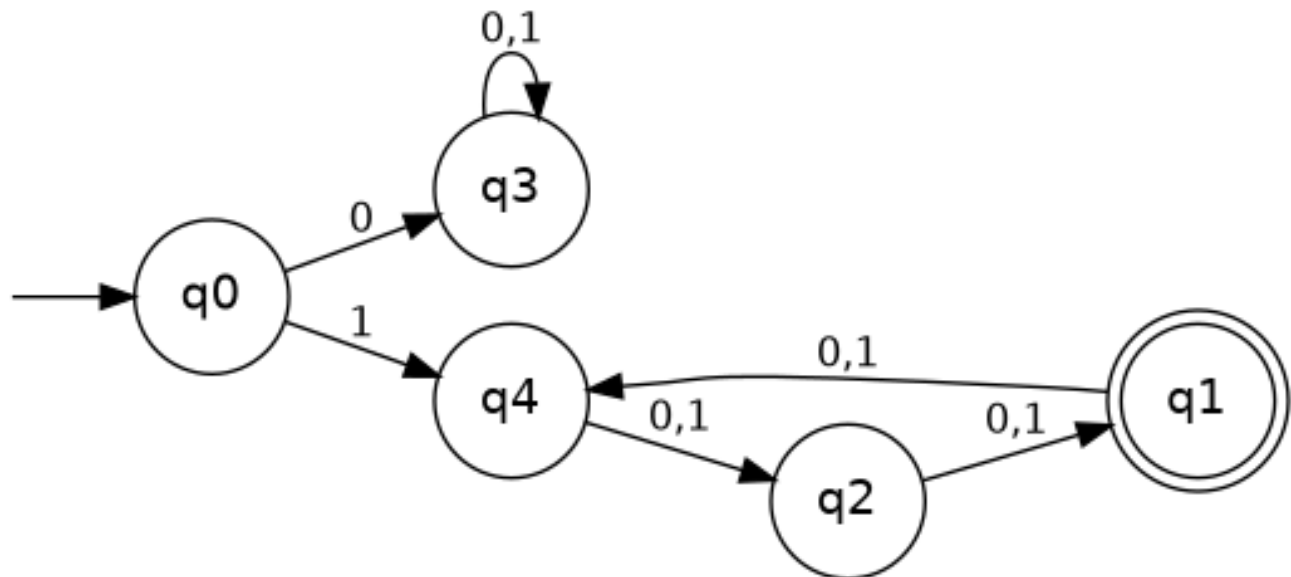
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q4	→	
q1	q4	q4		✓
q2	q1	q1		
q3	q3	q3		
q4	q2	q2		

State diagram (minimal DFA)



Question 75

Language: Exactly two 0s and ending with 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

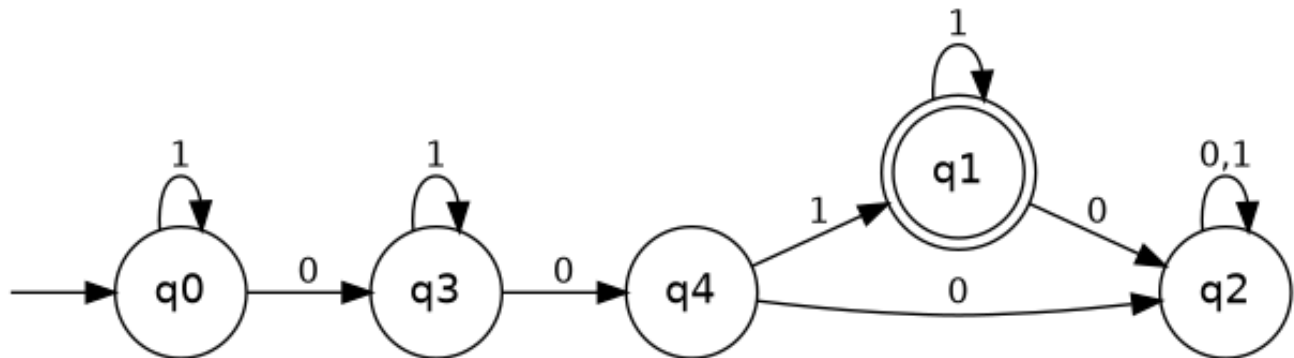
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q2	q1		✓
q2	q2	q2		
q3	q4	q3		
q4	q2	q1		

State diagram (minimal DFA)



Question 76

Language: Contains 101 and ends with 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

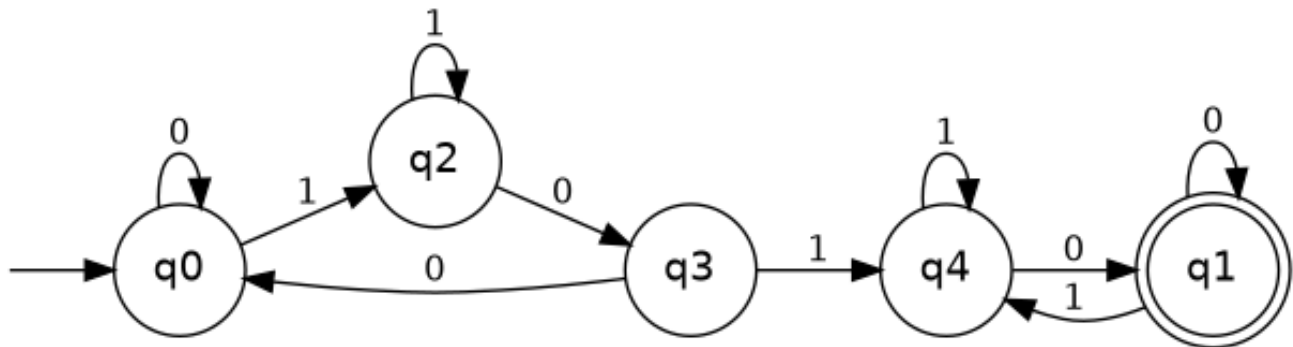
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q2	→	
q1	q1	q4		✓
q2	q3	q2		
q3	q0	q4		
q4	q1	q4		

State diagram (minimal DFA)



Question 77

Language: Every 0 is immediately followed by 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

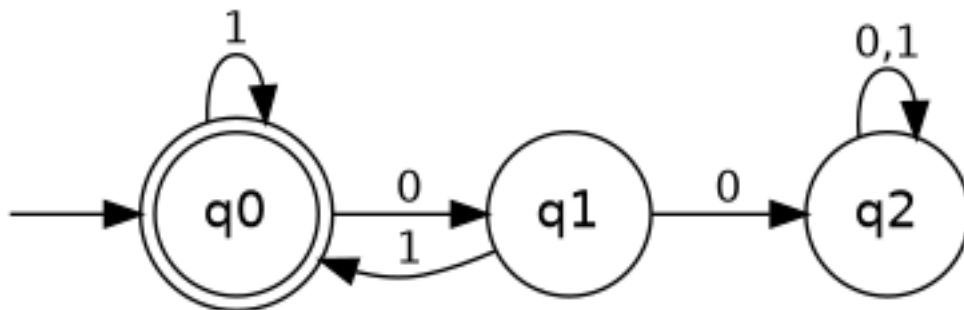
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		
q2	q2	q2		

State diagram (minimal DFA)



Question 78

Language: Every 1 is immediately preceded by 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

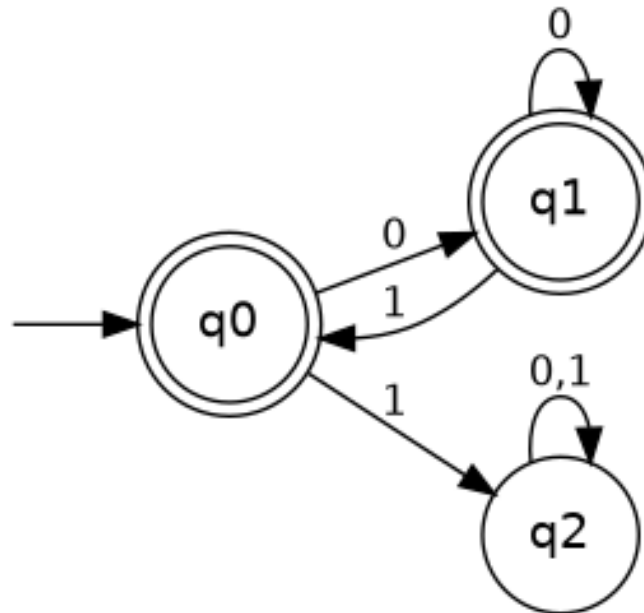
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q2	→	✓
q1	q1	q0		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 79

Language: No two consecutive 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

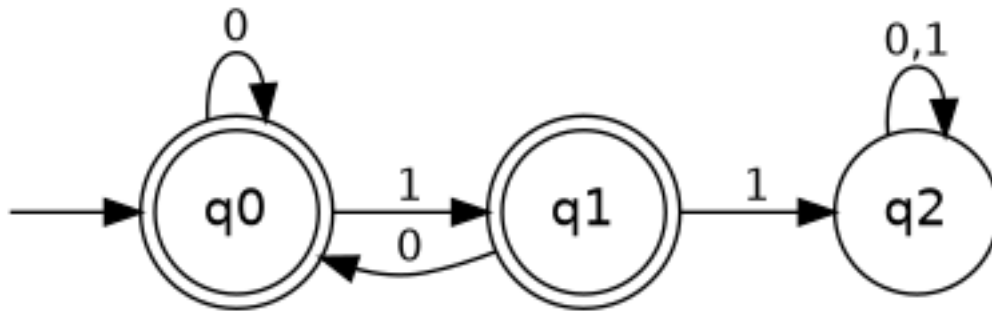
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q0	q2		✓
q2	q2	q2		

State diagram (minimal DFA)



Question 80

Language: No three consecutive 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

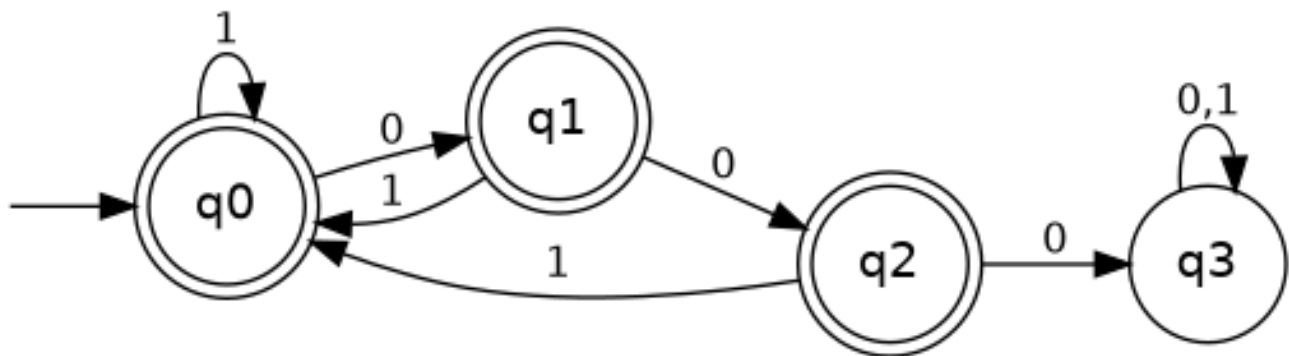
$q_0 = q_0$

$F = \{ q_0, q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		✓
q2	q3	q0		✓
q3	q3	q3		

State diagram (minimal DFA)



Question 81

Language: Third symbol from end is 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7 \}$ ($|Q| = 8$)

$\Sigma = \{ 0, 1 \}$

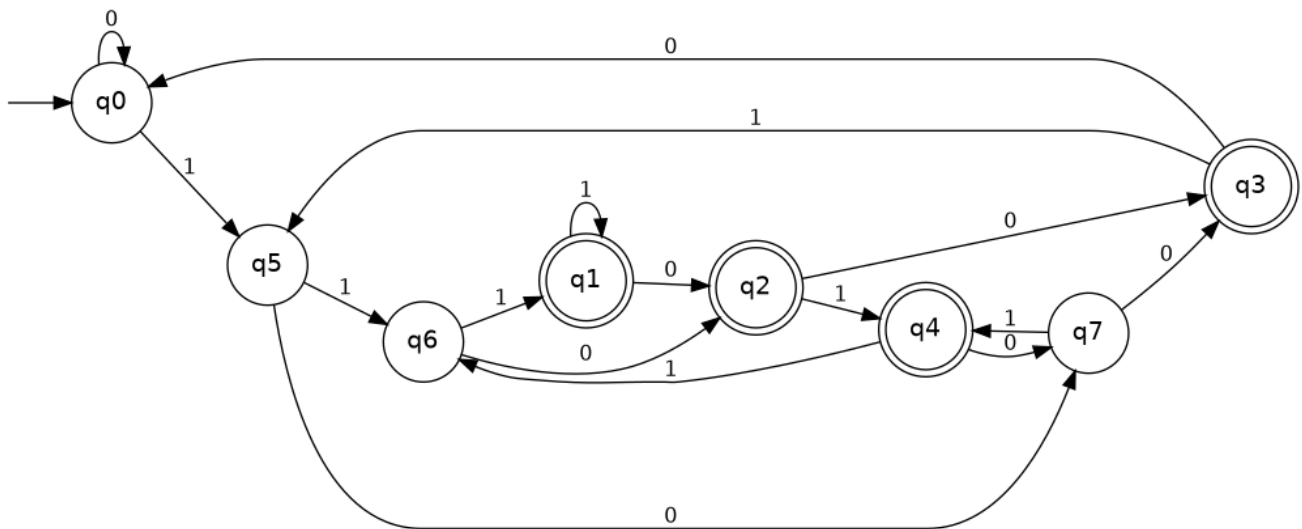
$q_0 = q_0$

$F = \{ q_1, q_2, q_3, q_4 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q7	→	
q1	q4	q3		✓
q2	q1	q2		✓
q3	q5	q6		✓
q4	q0	q7		✓
q5	q4	q3		
q6	q1	q2		
q7	q5	q6		

State diagram (minimal DFA)



Question 82

Language: Second symbol from end is 0.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

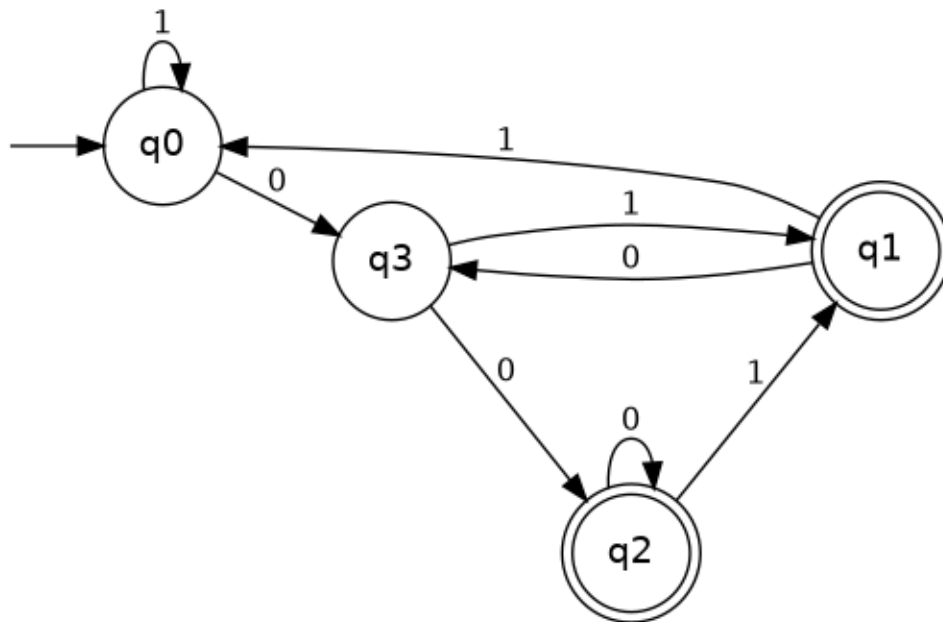
$q_0 = q_0$

$F = \{ q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q3	q0		✓
q2	q2	q1		✓
q3	q2	q1		

State diagram (minimal DFA)



Question 83

Language: Fourth symbol from end is 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15} \} \quad (|Q| = 16)$

$\Sigma = \{ 0, 1 \}$

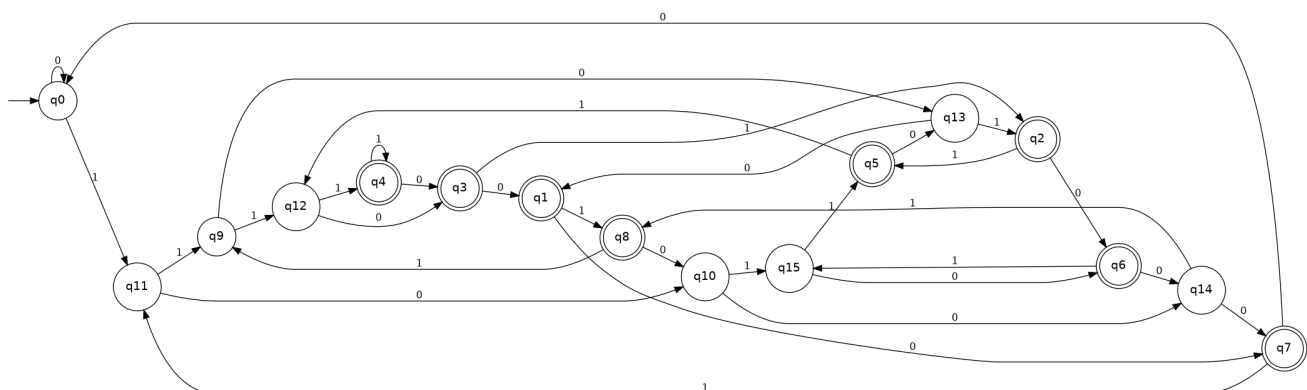
$q_0 = q_0$

$F = \{ q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q9	→	
q1	q14	q15		✓
q2	q12	q13		✓
q3	q11	q10		✓
q4	q0	q9		✓
q5	q2	q1		✓
q6	q4	q3		✓
q7	q8	q7		✓
q8	q6	q5		✓
q9	q11	q10		
q10	q14	q15		
q11	q12	q13		
q12	q4	q3		
q13	q2	q1		
q14	q6	q5		
q15	q8	q7		

State diagram (minimal DFA)



Question 84

Language: Number of 0s divisible by 3.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$ ($|Q| = 3$)

$\Sigma = \{ 0, 1 \}$

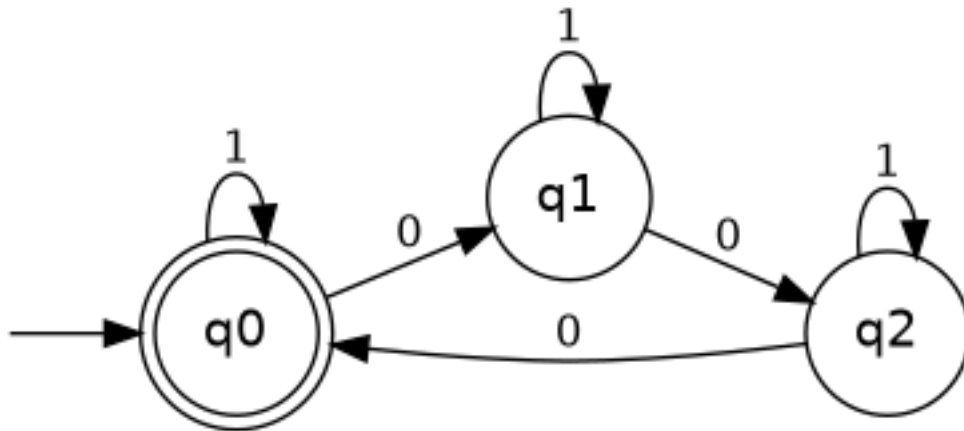
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q1		
q2	q0	q2		

State diagram (minimal DFA)



Question 85

Language: Number of 1s divisible by 4.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

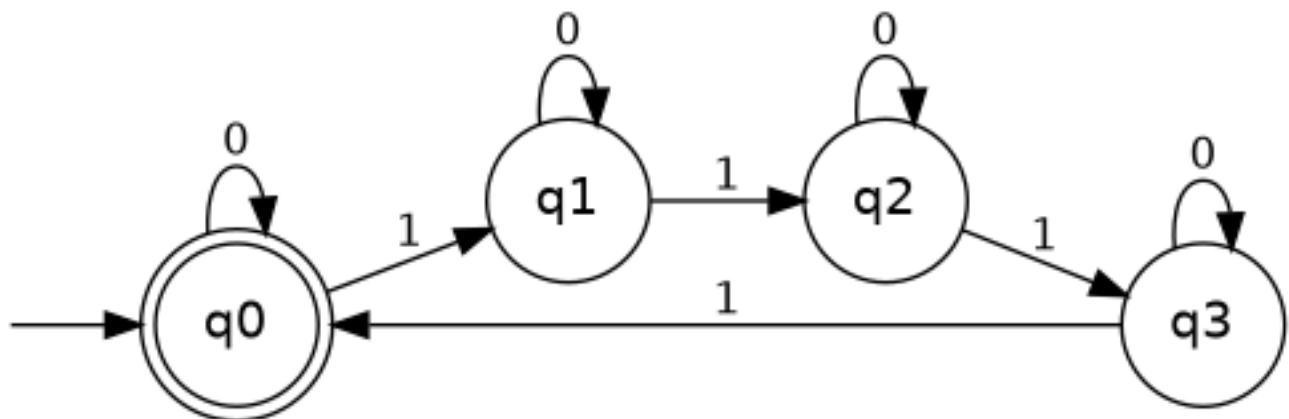
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q1	→	✓
q1	q1	q2		
q2	q2	q3		
q3	q3	q0		

State diagram (minimal DFA)



Question 86

Language: Both number of 0s and length even.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \} \quad (|Q| = 4)$

$\Sigma = \{ 0, 1 \}$

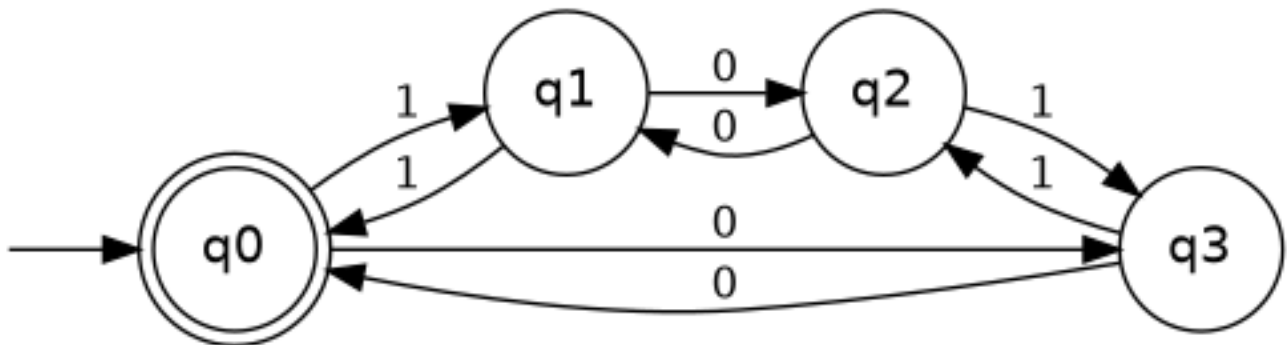
$q_0 = q_0$

$F = \{ q_0 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q1	→	✓
q1	q2	q0		
q2	q1	q3		
q3	q0	q2		

State diagram (minimal DFA)



Question 87

Language: Starts and ends with same symbol.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

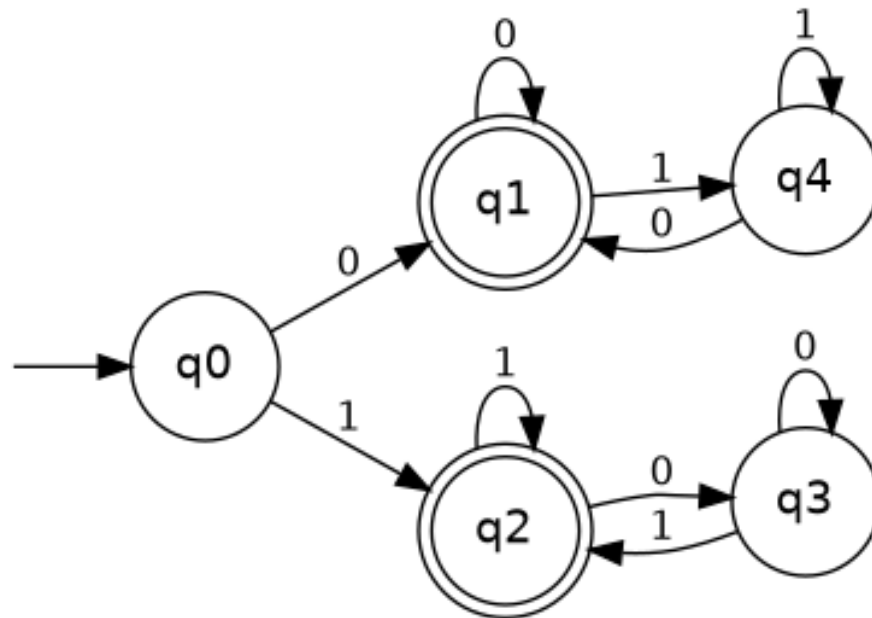
$q_0 = q_0$

$F = \{ q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q2	→	
q1	q1	q4		✓
q2	q3	q2		✓
q3	q3	q2		
q4	q1	q4		

State diagram (minimal DFA)



Question 88

Language: Starts with 01 and ends with 10.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

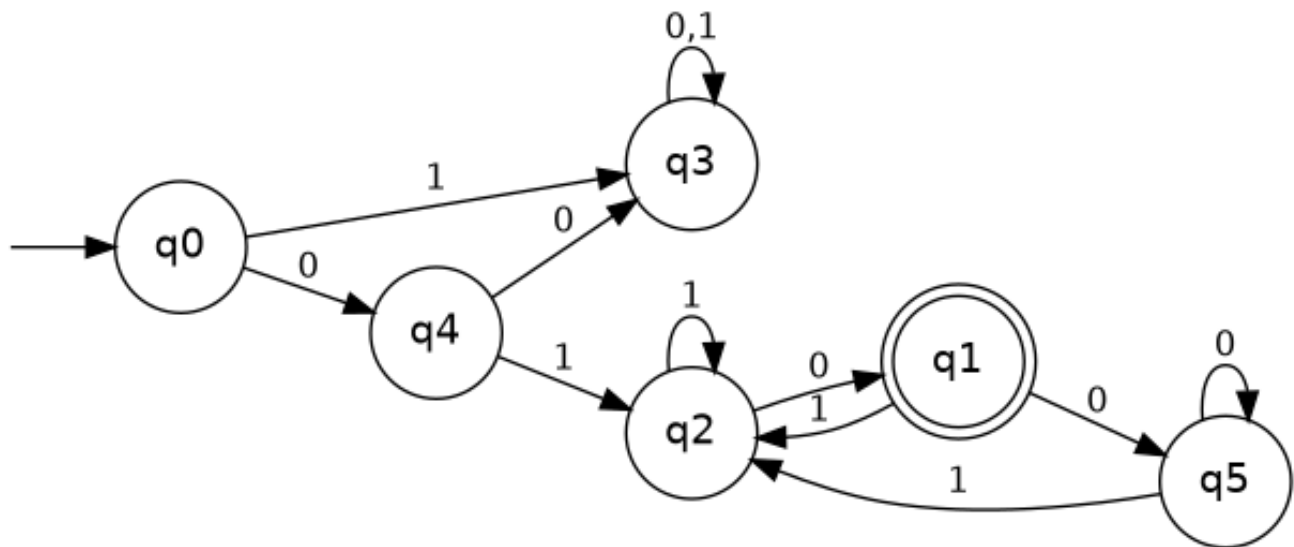
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q4	q3	→	
q1	q5	q2		✓
q2	q1	q2		
q3	q3	q3		
q4	q3	q2		
q5	q5	q2		

State diagram (minimal DFA)



Question 89

Language: Exactly one occurrence of 101.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6 \}$ ($|Q| = 7$)

$\Sigma = \{ 0, 1 \}$

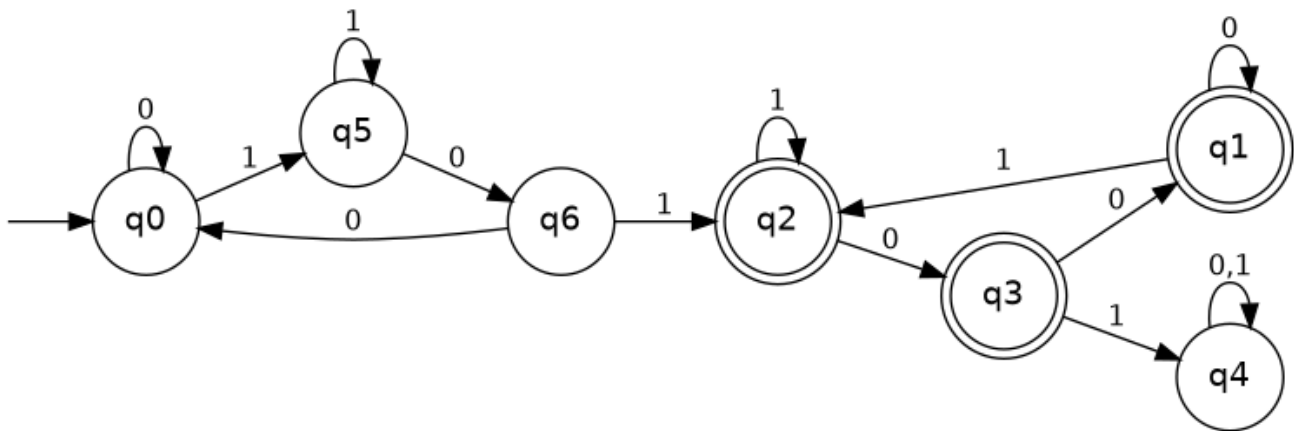
$q_0 = q_0$

$F = \{ q_1, q_2, q_3 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q5	→	
q1	q1	q2		✓
q2	q3	q2		✓
q3	q1	q4		✓
q4	q4	q4		
q5	q6	q5		
q6	q0	q2		

State diagram (minimal DFA)



Question 90

Language: At least two occurrences of 01.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \}$ ($|Q| = 5$)

$\Sigma = \{ 0, 1 \}$

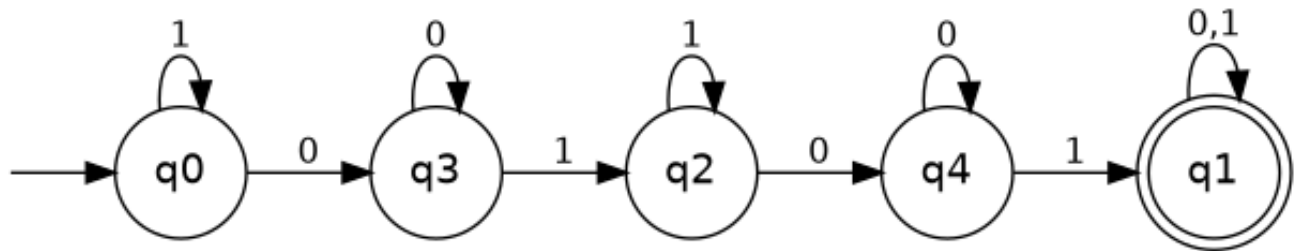
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q1	q1		✓
q2	q4	q2		
q3	q3	q2		
q4	q4	q1		

State diagram (minimal DFA)



Question 91

Language: Odd occurrences of 11.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

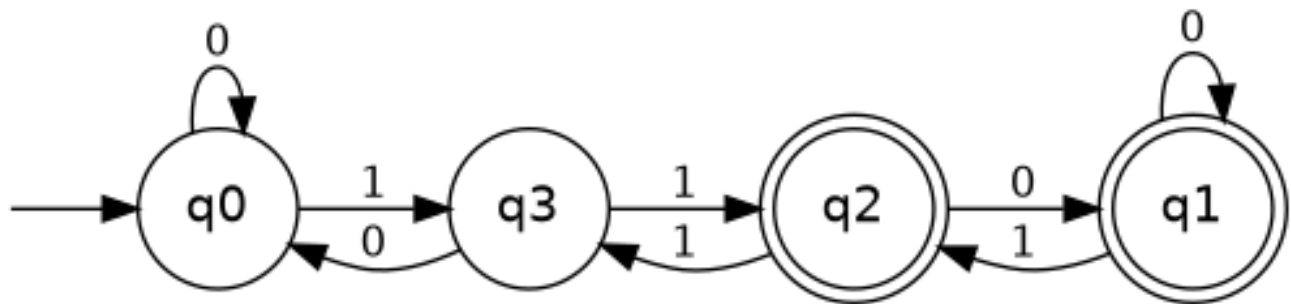
$q_0 = q_0$

$F = \{ q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q3	→	
q1	q1	q2		✓
q2	q1	q3		✓
q3	q0	q2		

State diagram (minimal DFA)



Question 92

Language: Does not contain 000 or 111.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

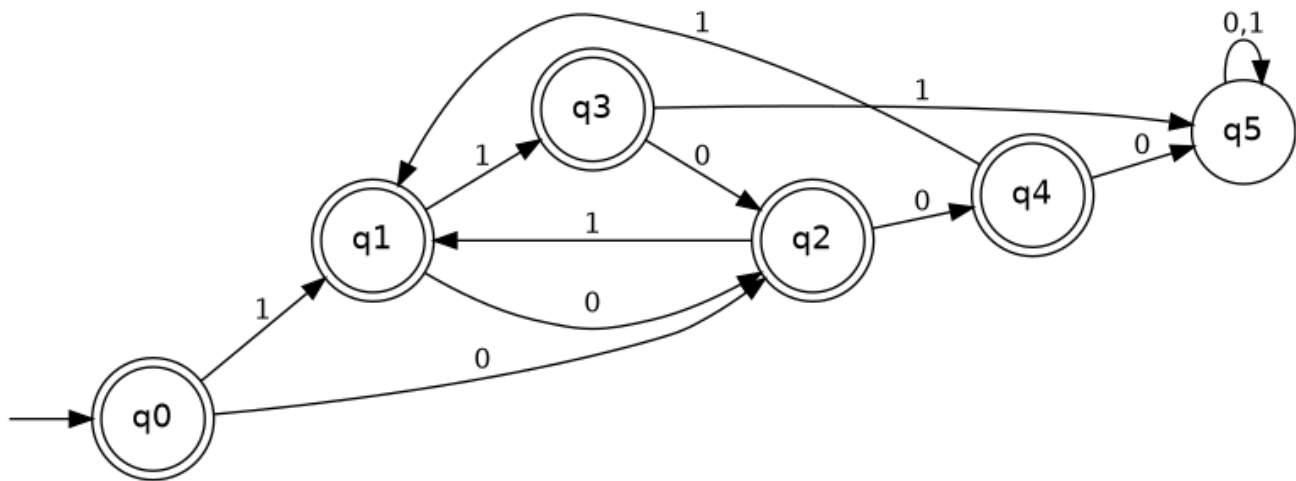
$q_0 = q_0$

$F = \{ q_0, q_1, q_2, q_3, q_4 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q1	→	✓
q1	q2	q3		✓
q2	q4	q1		✓
q3	q2	q5		✓
q4	q5	q1		✓
q5	q5	q5		

State diagram (minimal DFA)



Question 93

Language: Ends with 0011.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \}$ ($|Q| = 5$)

$\Sigma = \{ 0, 1 \}$

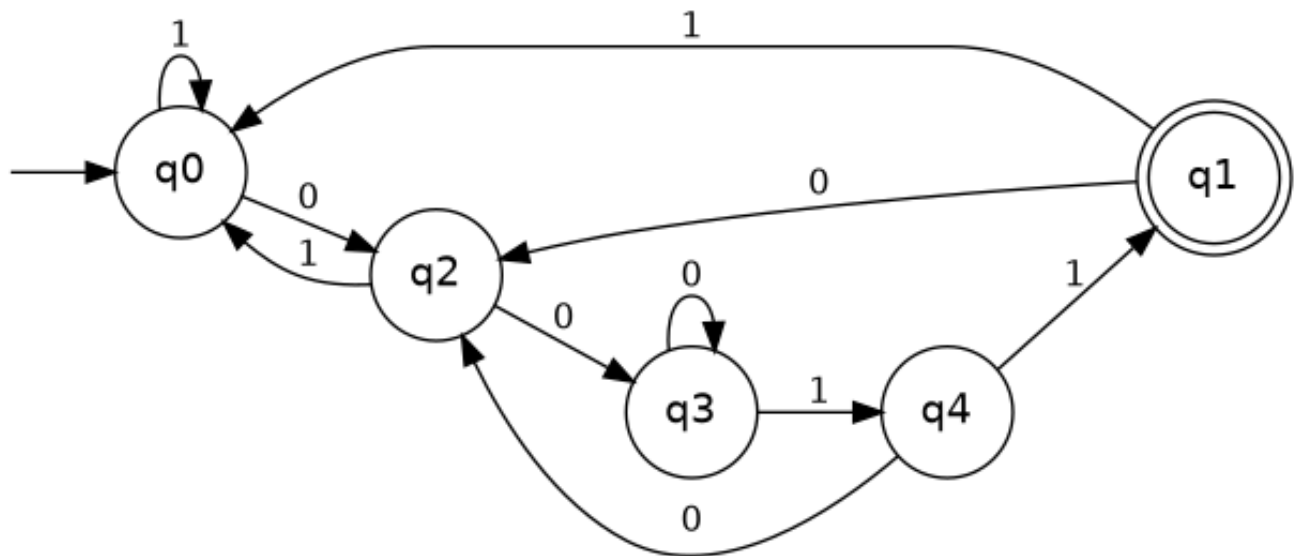
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q0	→	
q1	q2	q0		✓
q2	q3	q0		
q3	q3	q4		
q4	q2	q1		

State diagram (minimal DFA)



Question 94

Language: Begins with 1100.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \} \quad (|Q| = 6)$

$\Sigma = \{ 0, 1 \}$

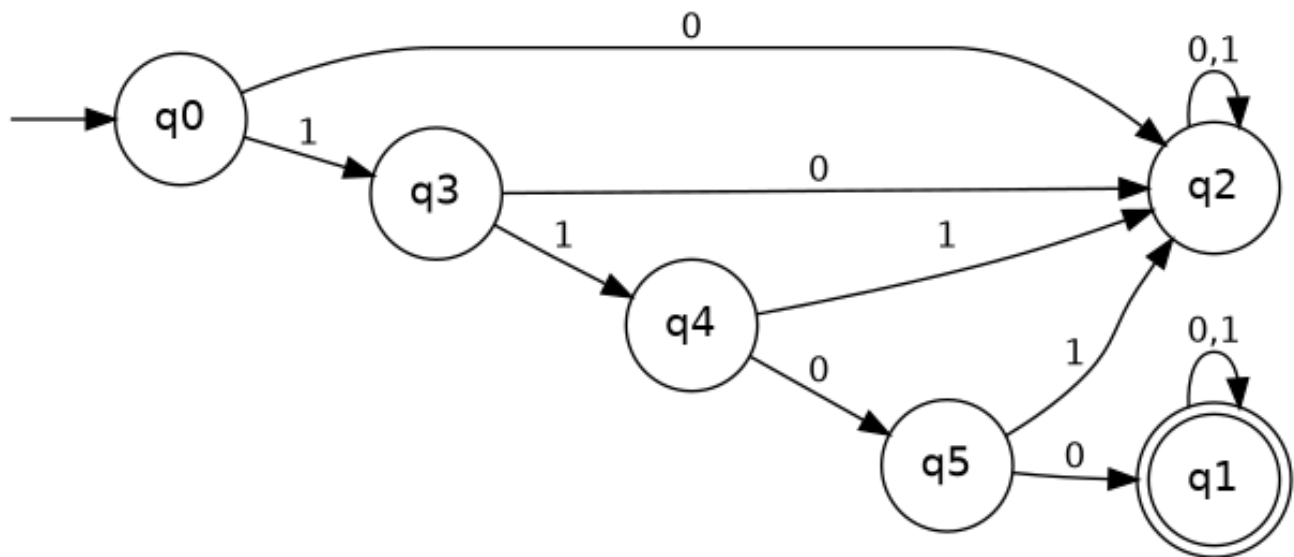
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q3	→	
q1	q1	q1		✓
q2	q2	q2		
q3	q2	q4		
q4	q5	q2		
q5	q1	q2		

State diagram (minimal DFA)



Question 95

Language: Contains both 101 and 010 (as substrings).

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11} \}$ ($|Q| = 12$)

$\Sigma = \{ 0, 1 \}$

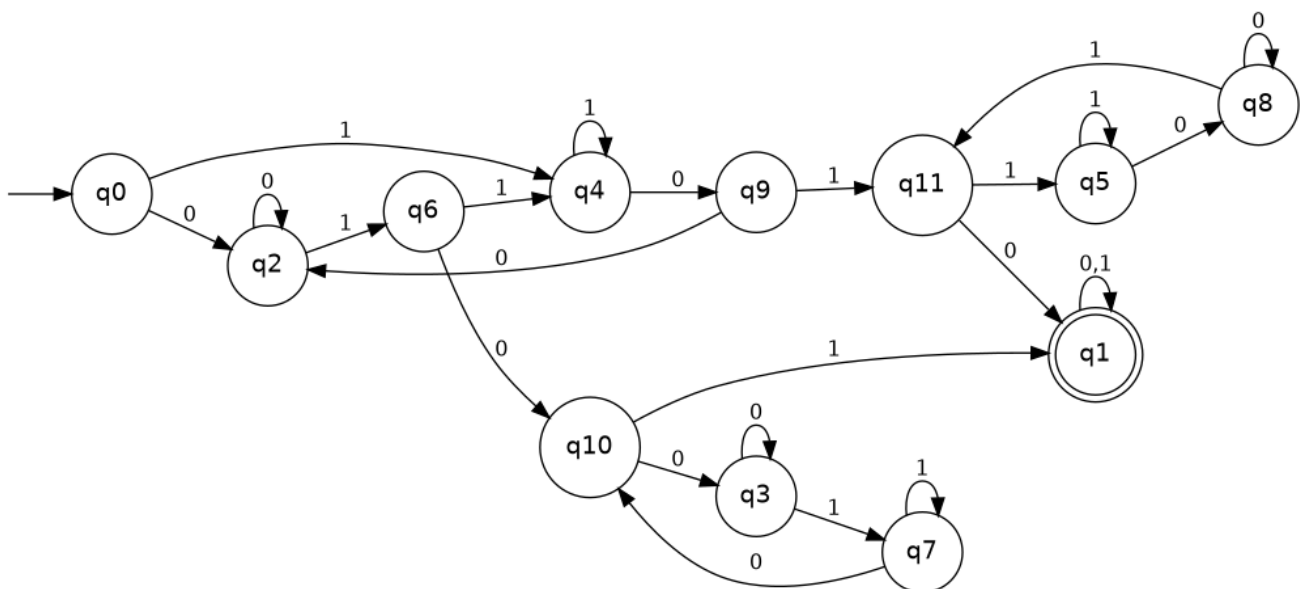
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q2	q4	→	
q1	q1	q1		✓
q2	q2	q6		
q3	q3	q7		
q4	q9	q4		
q5	q8	q5		
q6	q10	q4		
q7	q10	q7		
q8	q8	q11		
q9	q2	q11		
q10	q3	q1		
q11	q1	q5		

State diagram (minimal DFA)



Question 96

Language: Exactly one pair of consecutive 1s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

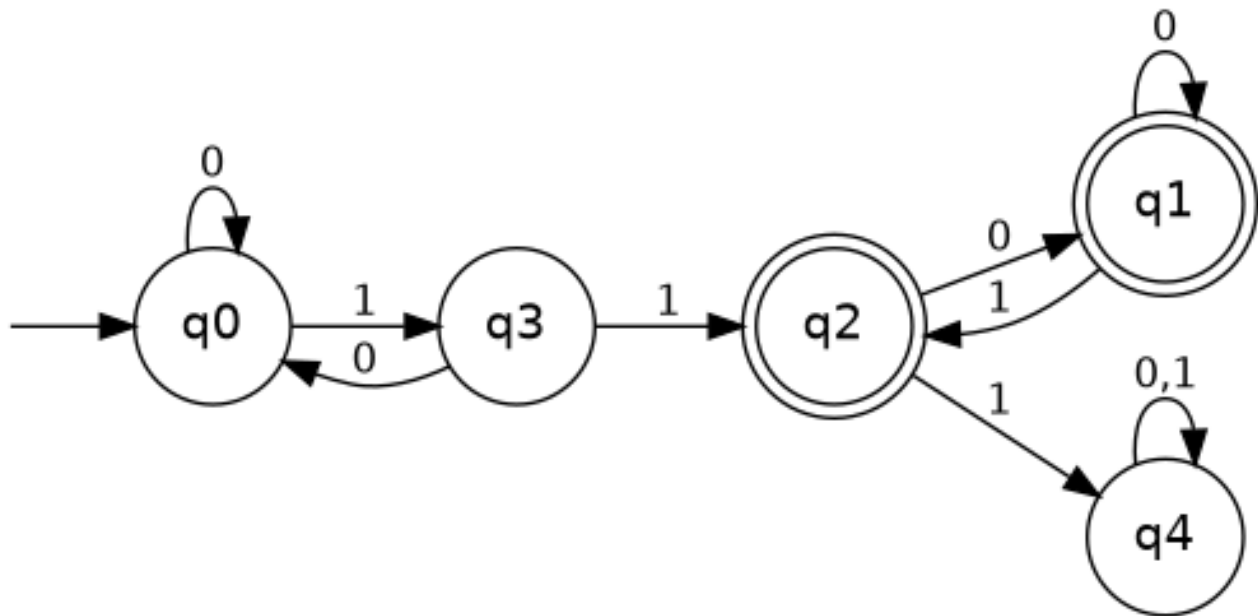
$q_0 = q_0$

$F = \{ q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q0	q3	→	
q1	q1	q2		✓
q2	q1	q4		✓
q3	q0	q2		
q4	q4	q4		

State diagram (minimal DFA)



Question 97

Language: Exactly one pair of consecutive 0s.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \} \quad (|Q| = 5)$

$\Sigma = \{ 0, 1 \}$

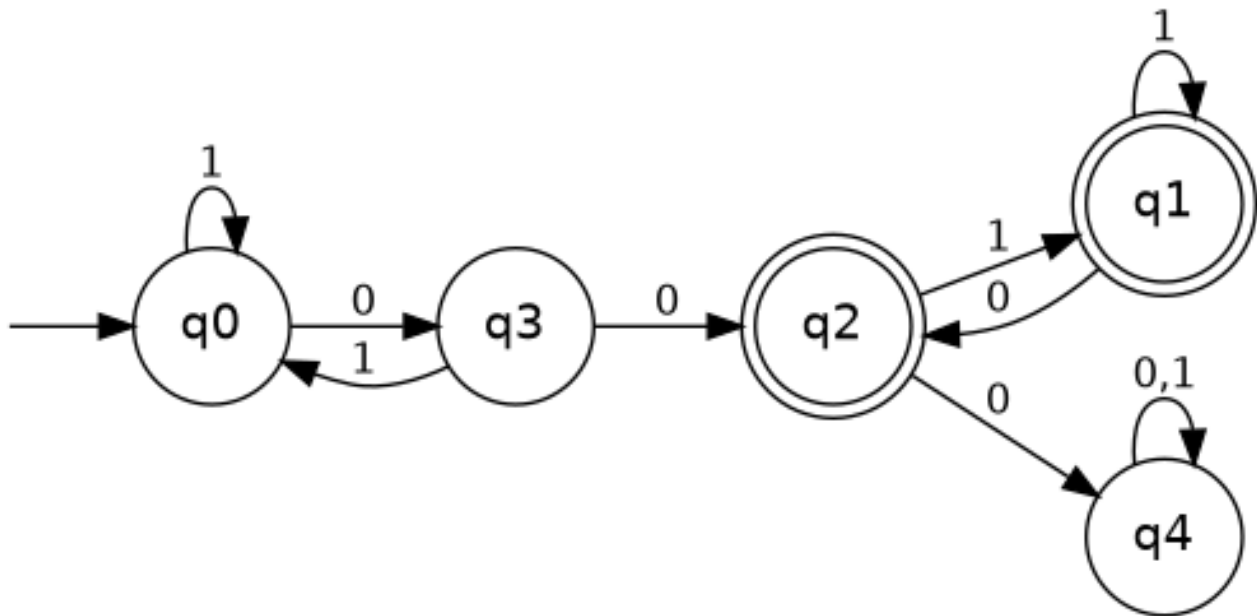
$q_0 = q_0$

$F = \{ q_1, q_2 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q0	→	
q1	q2	q1		✓
q2	q4	q1		✓
q3	q2	q0		
q4	q4	q4		

State diagram (minimal DFA)



Question 98

Language: At least one 0 and one 1.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

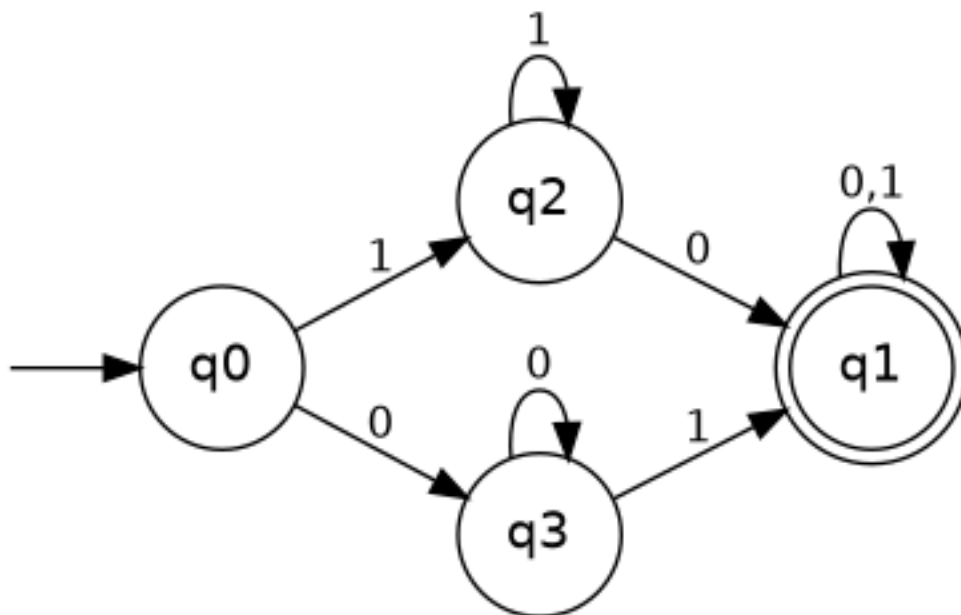
$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q3	q2	→	
q1	q1	q1		✓
q2	q1	q2		
q3	q3	q1		

State diagram (minimal DFA)



Question 99

Language: Every 00 is followed later by 11.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3 \}$ ($|Q| = 4$)

$\Sigma = \{ 0, 1 \}$

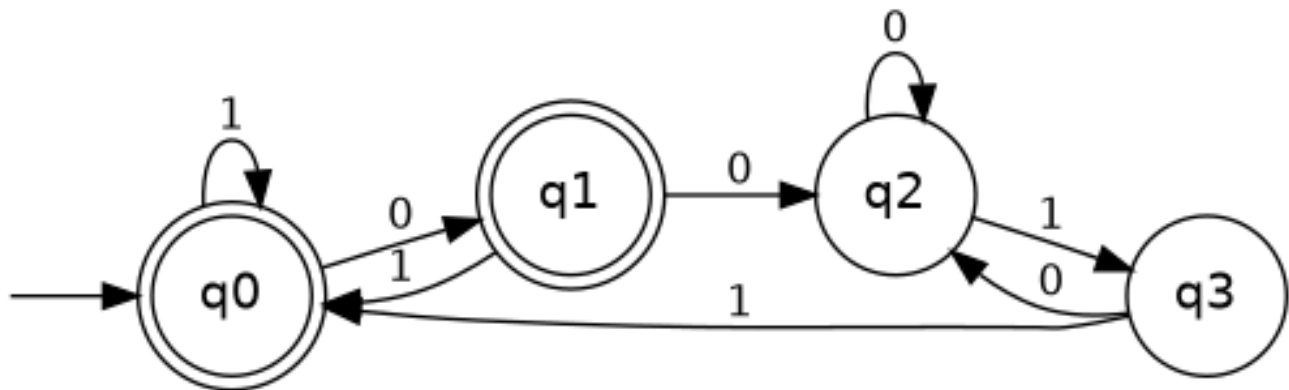
$q_0 = q_0$

$F = \{ q_0, q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q0	q1	q0	→	✓
q1	q2	q0		✓
q2	q2	q3		
q3	q2	q0		

State diagram (minimal DFA)



Question 100

Language: All strings except epsilon.

Formal definition

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1 \}$ ($|Q| = 2$)

$\Sigma = \{ 0, 1 \}$

$q_0 = q_0$

$F = \{ q_1 \}$

Transition table

State	on 0	on 1	Start?	Accept?
q_0	q_1	q_1	$\rightarrow$	
q_1	q_1	q_1		✓

State diagram (minimal DFA)

